

The Good, the Bad, and the Ugly Coronavirus Graphs

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Southwest Michigan Chapter of the
American Statistical Association (ASA),

Virtual

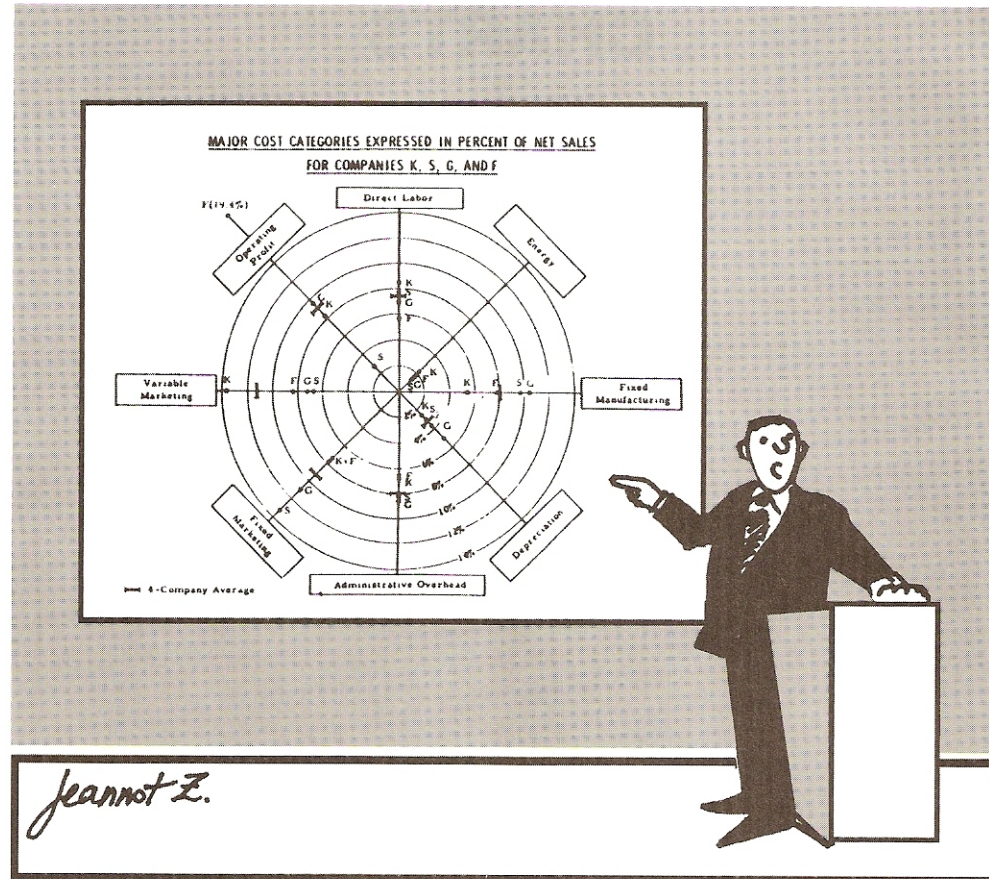
January 7, 2021



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Motivation



"What do you mean, what does it mean?"

From: Zelazny, G. (2001), Say it with Charts: The Executive's Guide to Visual Communication (Fourth Edition), McGraw-Hill, New York, NY.

Motivation

- As an expert in statistical graphics, it hurts to see bad graphs in the news, on the web, and produced by students.
- Even worse are bad graphs in science (including those from articles in peer-reviewed journals).
- As the former chair of a task force of the *Statistical Graphics Section of the American Statistical Association (ASA)*, we reevaluated the winning posters of the annual ASA poster competition for children from kindergarten to grade 12 – and noticed many bad graphs, even among the winners.
- I am permanently on the lookout for bad graphs.

“How to Display Data Badly”

“How to Display Data Badly”

- From: Wainer, H. (1997), Visual Revelations: Graphical Tales of Fate and Deception from Napoleon Bonaparte to Ross Perot, Copernicus/Springer, New York, NY:
- “The aim of good data graphics is to display data accurately and clearly. [...]
- Thus, if we wish to display data badly, we have three avenues to follow.
 - A. Don't show much data.
 - B. Show the data inaccurately.
 - C. Obfuscate the data.” [i.e., show the data unclearly]

A. Don't show much data

- Rule 1: Show as little data as possible (minimize the data density).
- Rule 2: Hide what data you do show (minimize the data/ink ratio).

B. Show the data inaccurately

- Rule 3: Ignore the visual metaphor altogether.
- Rule 4: Only order matters.
- Rule 5: Graph data out of context.

C. Obfuscate the data

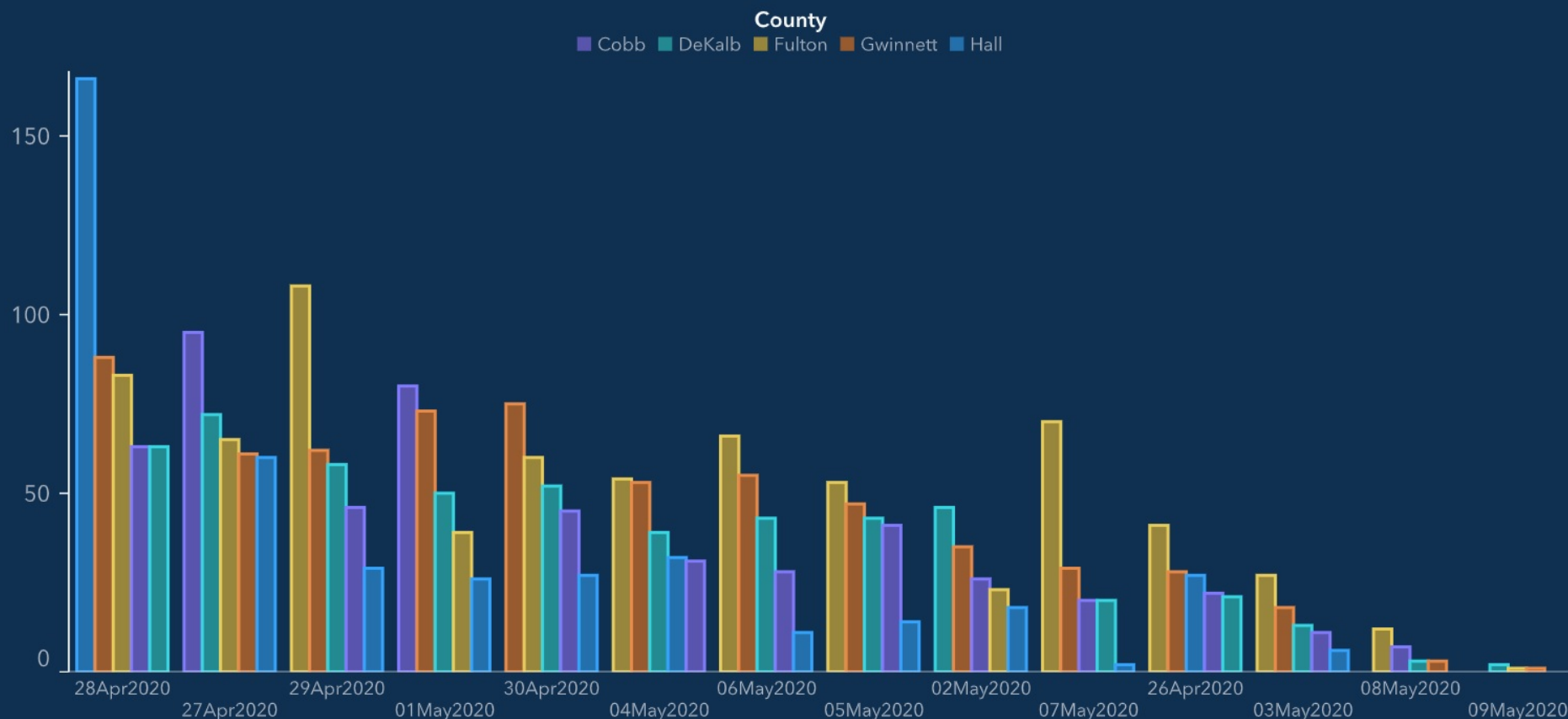
- Rule 6: Change scales in mid-axis.
- Rule 7: Emphasize the trivial (ignore the important).
- Rule 8: Jiggle the baseline.
- Rule 9: Alabama first!
- Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously.
- Rule 11: More is murkier: (a) more decimal places and (b) more dimensions.
- Rule 12: If it has been done well in the past, think of a new way to do it.

The Bad and the Ugly Ones

Georgia's Covid-19 Cases – The Original

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.

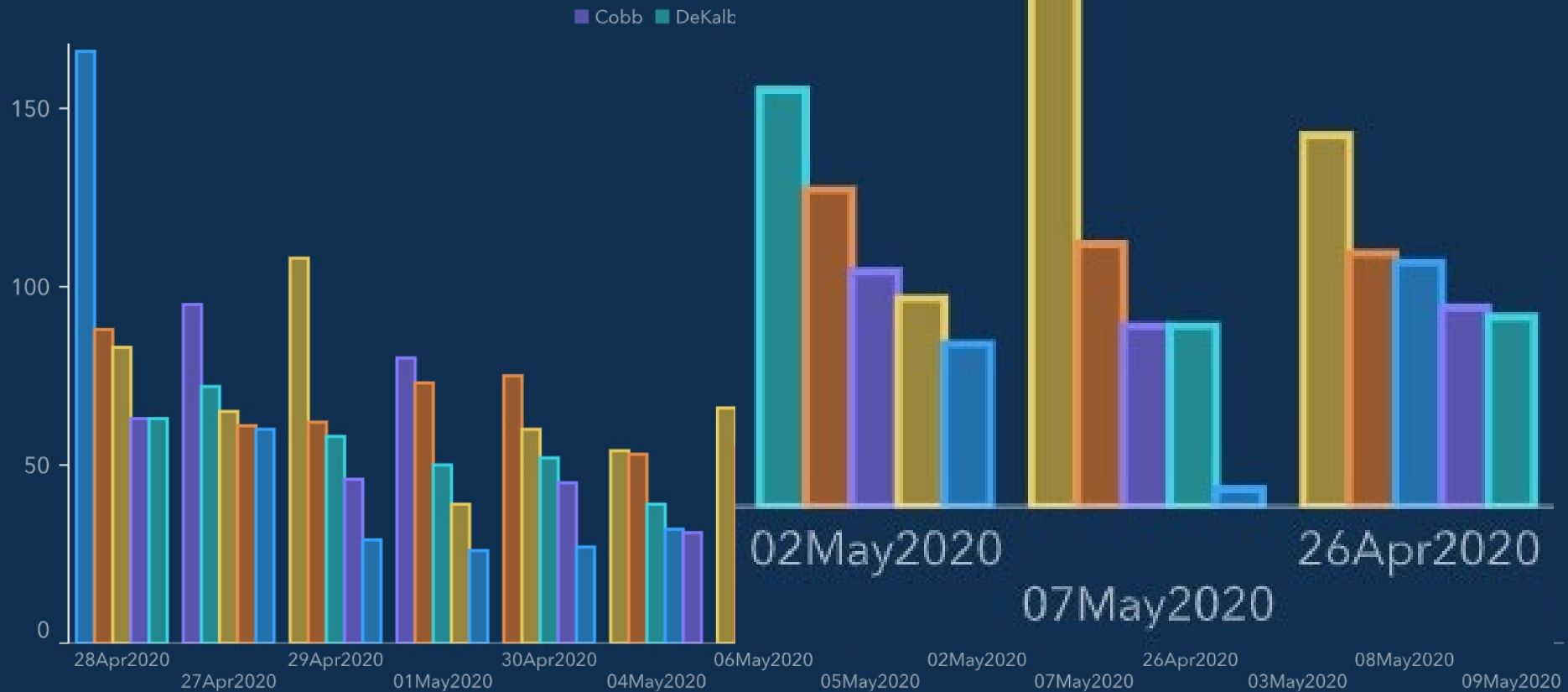


Georgia Department of Public Health (Graph no longer available from source)

Georgia's Covid-19 Cases – Close-Up

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

The chart below represents the most impacted counties over also represents the number of deaths and hospitalizations in €



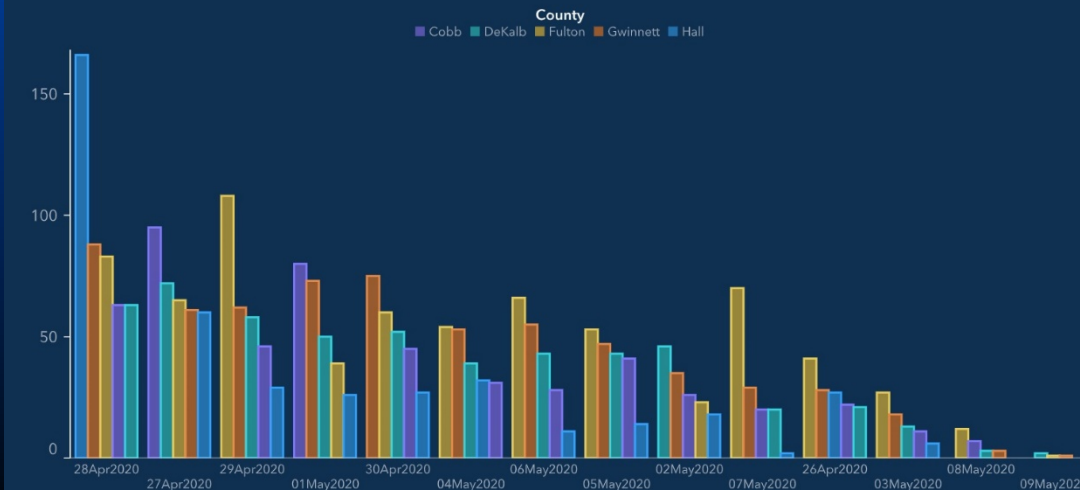
Georgia's Covid-19 Cases – Bad Graph

Rules Followed

- Rule 3: Ignore the visual metaphor altogether: Dates arbitrarily arranged on horizontal axis.
- Rule 7: Emphasize the trivial (ignore the important): Different ordering of counties in each block of five, making it hard to compare / follow a county on various dates.

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

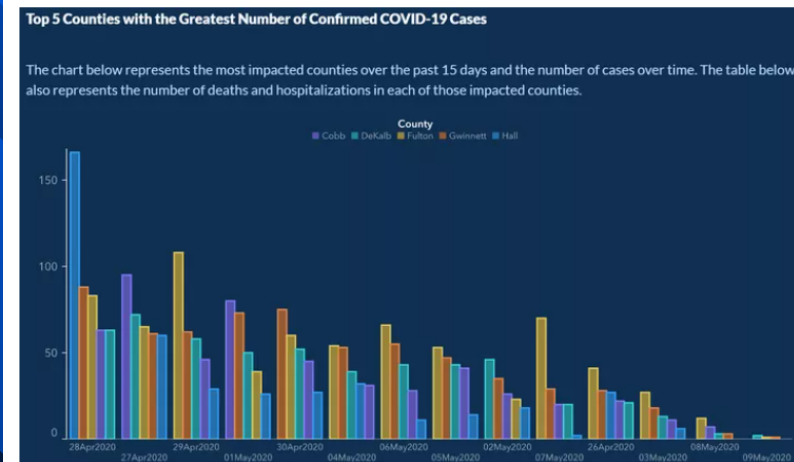
The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.



Georgia's Covid-19 Cases – Discussion (1)



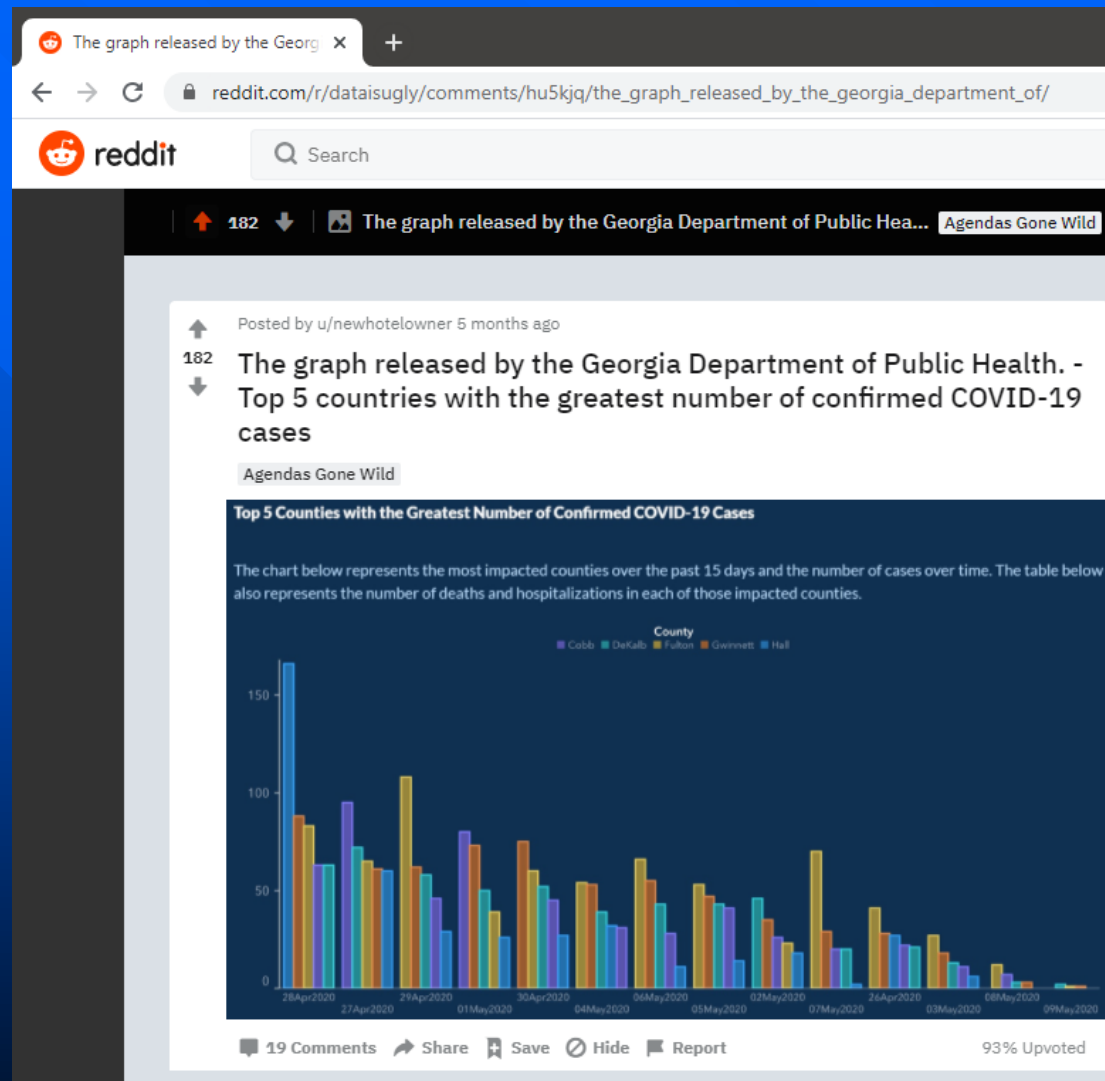
Georgia's Department of Public Health has been criticized in particular for sharing misleading data. Last week, it corrected a graph on its website that appeared to show confirmed case counts decreasing (shown below); the data, however, was not arranged in chronological order. The x-axis of the graph was not ordered chronologically, a method that caused the highest values to be clustered on the left, and the lowest on the right, regardless of when those values were recorded.



A Georgia Department of Public Health graphic prior to the correction.

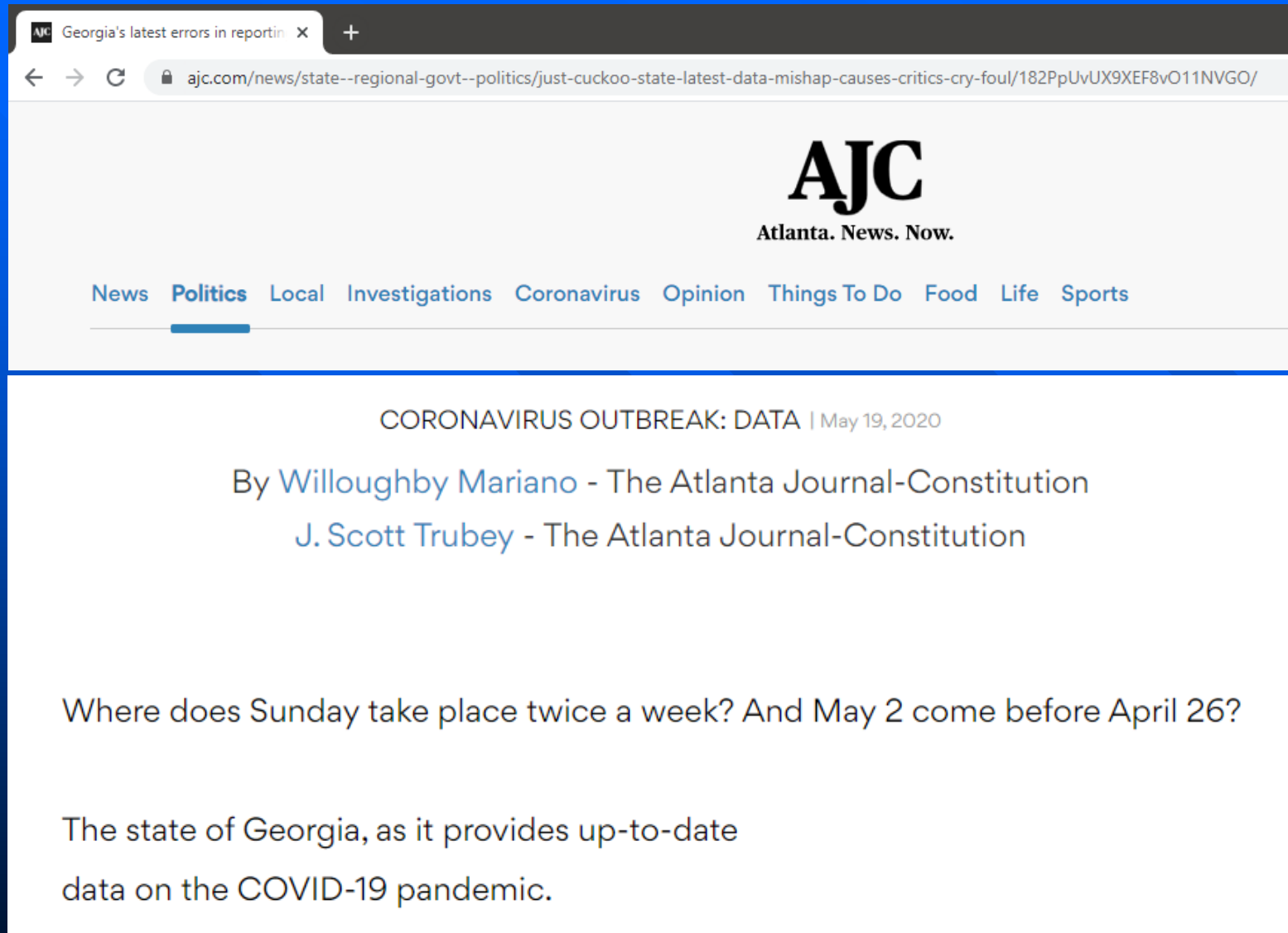
<https://www.vox.com/covid-19-coronavirus-us-response-trump/2020/5/18/21262265/georgia-covid-19-cases-declining-reopening>

Georgia's Covid-19 Cases – Discussion (2)



https://www.reddit.com/r/dataisugly/comments/hu5kjq/the_graph_released_by_the_georgia_department_of/

Georgia's Covid-19 Cases – Discussion (3)



The image is a screenshot of a web browser displaying a news article from the Atlanta Journal-Constitution (AJC). The browser's address bar shows the URL: [ajc.com/news/state--regional-govt--politics/just-cuckoo-state-latest-data-mishap-causes-critics-cry-foul/182PpUvUX9XEF8vO11NVGO/](https://www.ajc.com/news/state--regional-govt--politics/just-cuckoo-state-latest-data-mishap-causes-critics-cry-foul/182PpUvUX9XEF8vO11NVGO/). The AJC logo is prominently displayed in the center, with the tagline "Atlanta. News. Now." below it. A navigation bar at the top lists various news categories: News, Politics (which is highlighted with a blue underline), Local, Investigations, Coronavirus, Opinion, Things To Do, Food, Life, and Sports. The article's title, "CORONAVIRUS OUTBREAK: DATA | May 19, 2020", is centered above the byline. The byline lists two authors: "By Willoughby Mariano - The Atlanta Journal-Constitution" and "J. Scott Trubey - The Atlanta Journal-Constitution". The main text of the article begins with the sentence: "Where does Sunday take place twice a week? And May 2 come before April 26?" followed by another sentence: "The state of Georgia, as it provides up-to-date data on the COVID-19 pandemic."

Georgia's latest errors in reportin x +

← → ↺ [ajc.com/news/state--regional-govt--politics/just-cuckoo-state-latest-data-mishap-causes-critics-cry-foul/182PpUvUX9XEF8vO11NVGO/](https://www.ajc.com/news/state--regional-govt--politics/just-cuckoo-state-latest-data-mishap-causes-critics-cry-foul/182PpUvUX9XEF8vO11NVGO/)

AJC
Atlanta. News. Now.

News **Politics** Local Investigations Coronavirus Opinion Things To Do Food Life Sports

CORONAVIRUS OUTBREAK: DATA | May 19, 2020

By Willoughby Mariano - The Atlanta Journal-Constitution
J. Scott Trubey - The Atlanta Journal-Constitution

Where does Sunday take place twice a week? And May 2 come before April 26?

The state of Georgia, as it provides up-to-date data on the COVID-19 pandemic.

Georgia's Covid-19 Cases – Discussion (4)

"It's just cuckoo," Democratic state Rep. Scott Holcomb of Atlanta told the Atlanta Journal-Constitution on May 13. "I don't know how anyone can defend this graph as not being misleading. I really don't."

In a Facebook post on May 11, Holcomb said he sent a letter to Republican Gov. Brian Kemp and "received a note back that the data was corrected to be arranged in chronological order."

"I have two master's degrees and have worked with a fair amount of data. I've never seen anything like this graph in my life," he wrote.

Georgia apologizes over 'processing error' after accusations of manipulating COVID-19 case counts

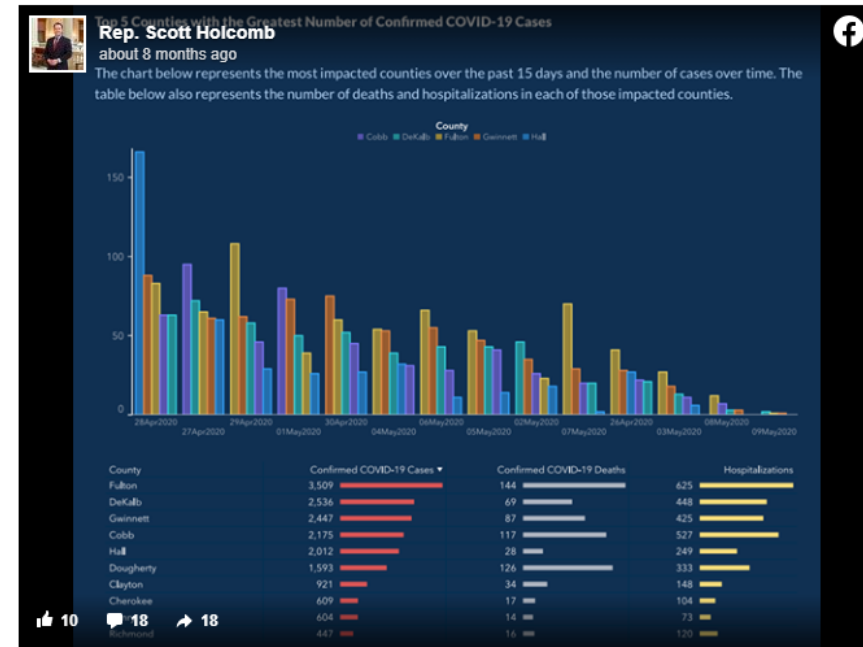
FOX5 ATLANTA

Live News Weather Good Day Sports Contests Email

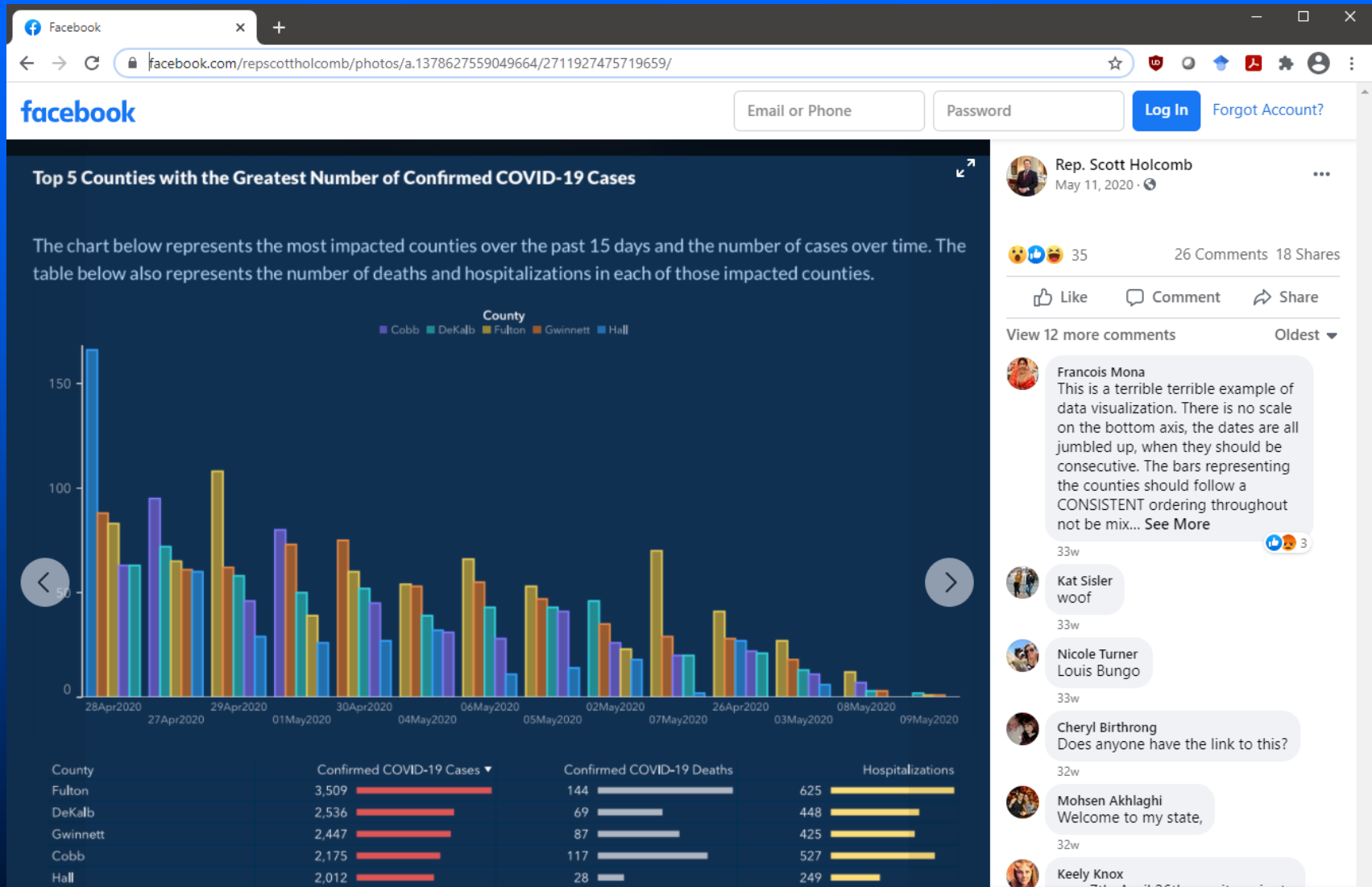
Special Weather Statement until WED 11:00 AM EST, Fannin County, Union County, Towns County

Georgia apologizes over 'processing error' after accusations of manipulating COVID-19 case counts

By Danielle Wallace | Published May 19, 2020 | Coronavirus in Georgia | FOX News

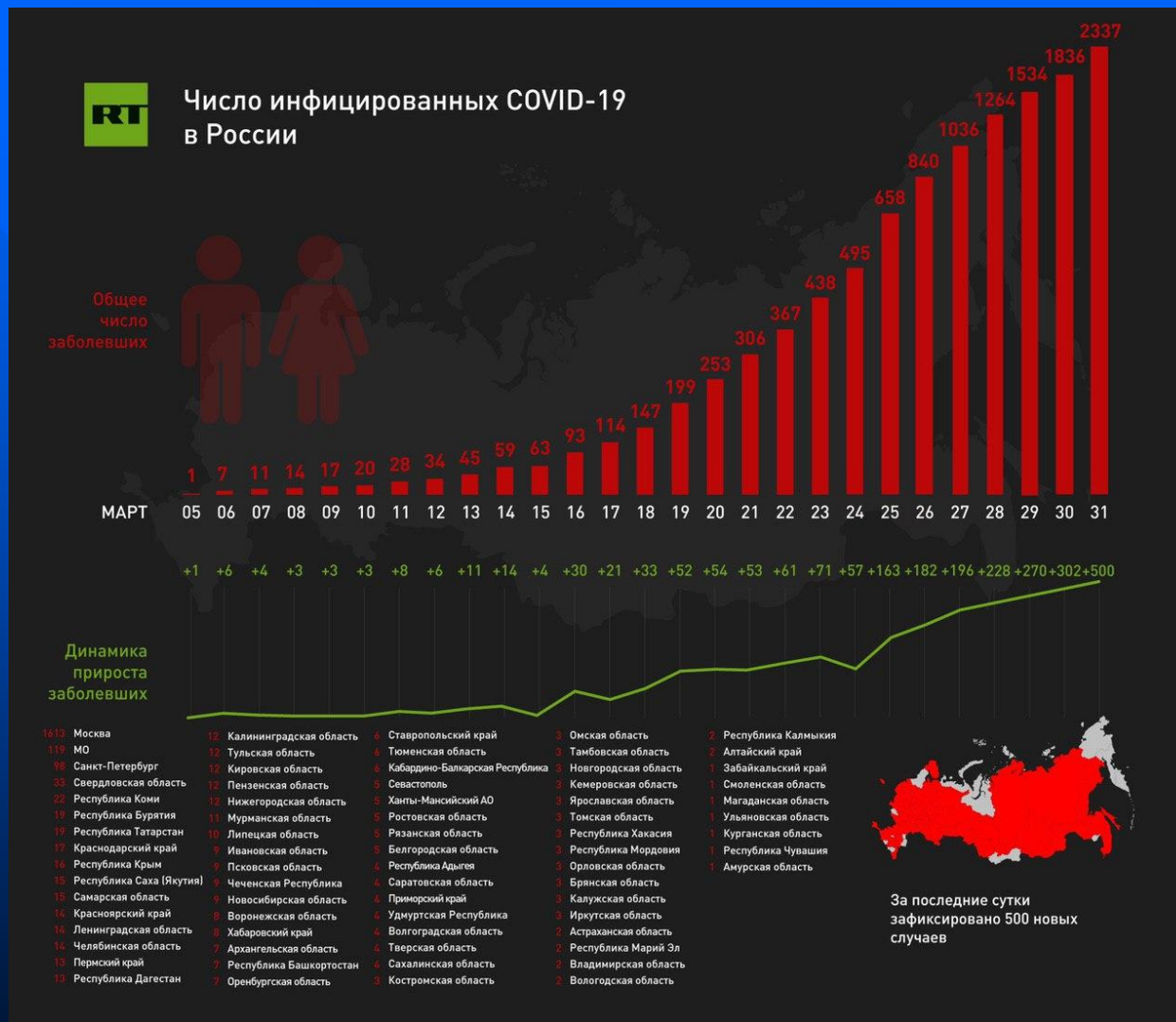


Georgia's Covid-19 Cases – Discussion (5)



<https://www.facebook.com/repscottholcomb/photos/a.1378627559049664/2711927475719659/>

Russian Covid-19 Cases – The Original



RT in Russian (Graph no longer available from source)

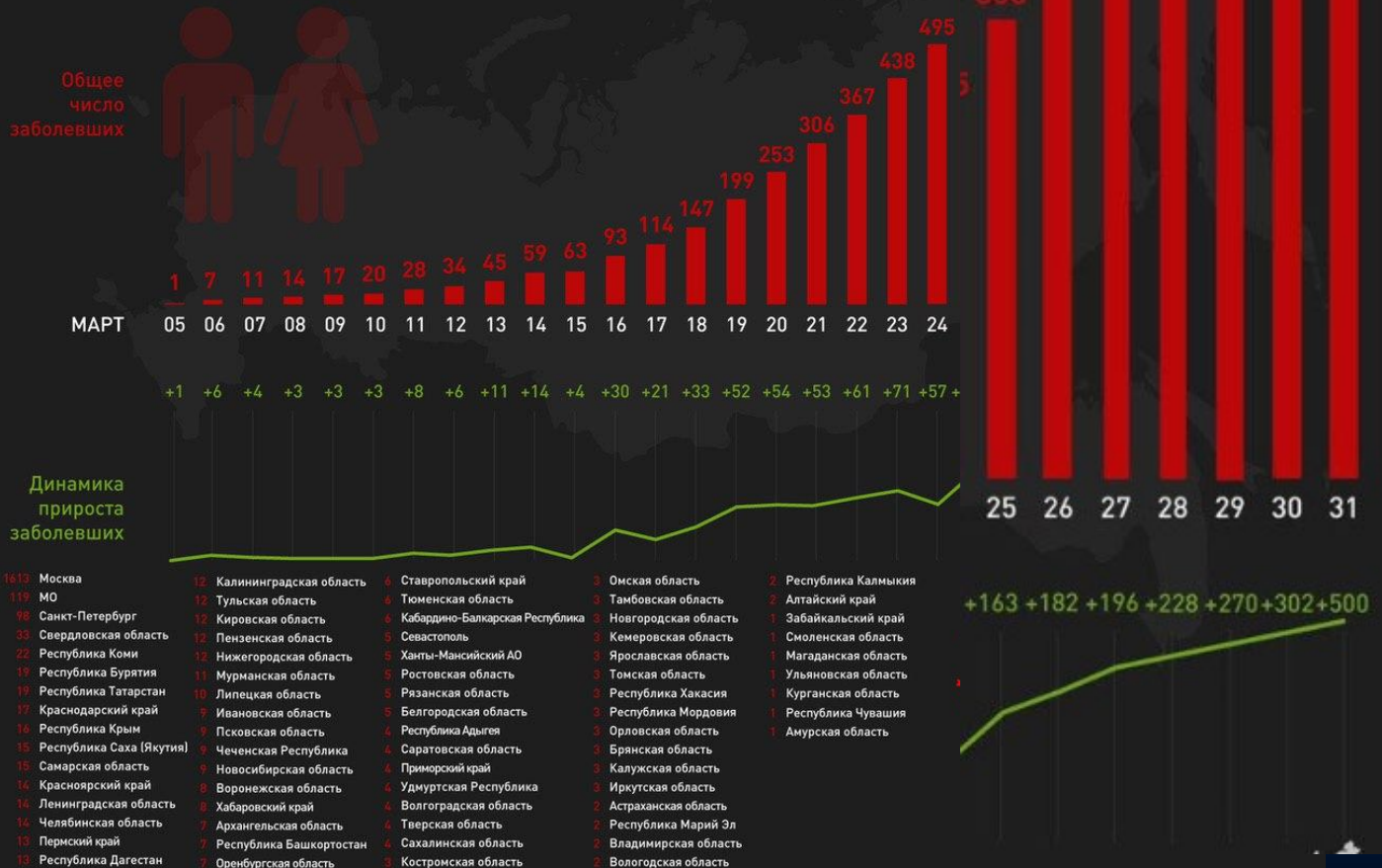
Russian Covid-19 Cases – Close-Up

Number of Covid-19 Infections in Russia

General Number of Sick People



Число инфицированных COVID-19
в России

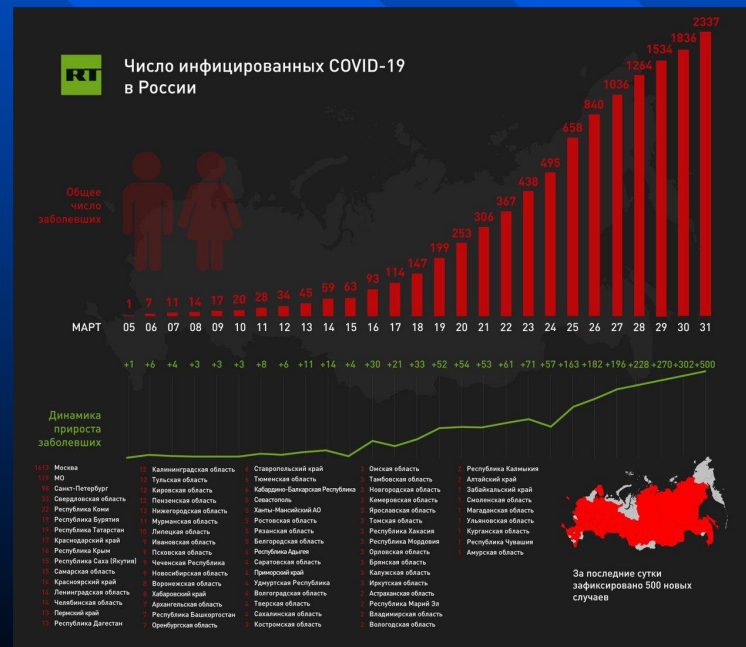


Dynamic of Growth of Sick People

Russian Covid-19 Cases – Bad Graph

Rules Followed

- Rule 4: Only order matters: Heights of red bars are not proportional to the numbers shown. Same for the green line.
- Rule 6: Change scales in mid-axis: In fact, no scale for the vertical axes are shown at all. Could these be log-scales?

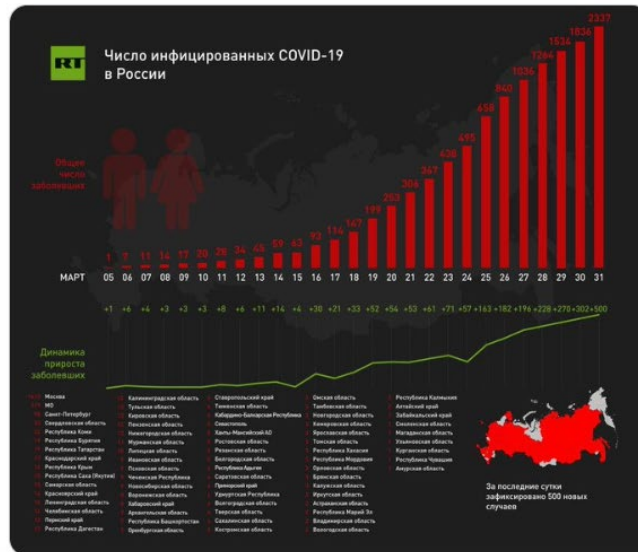


Russian Covid-19 Cases – Discussion

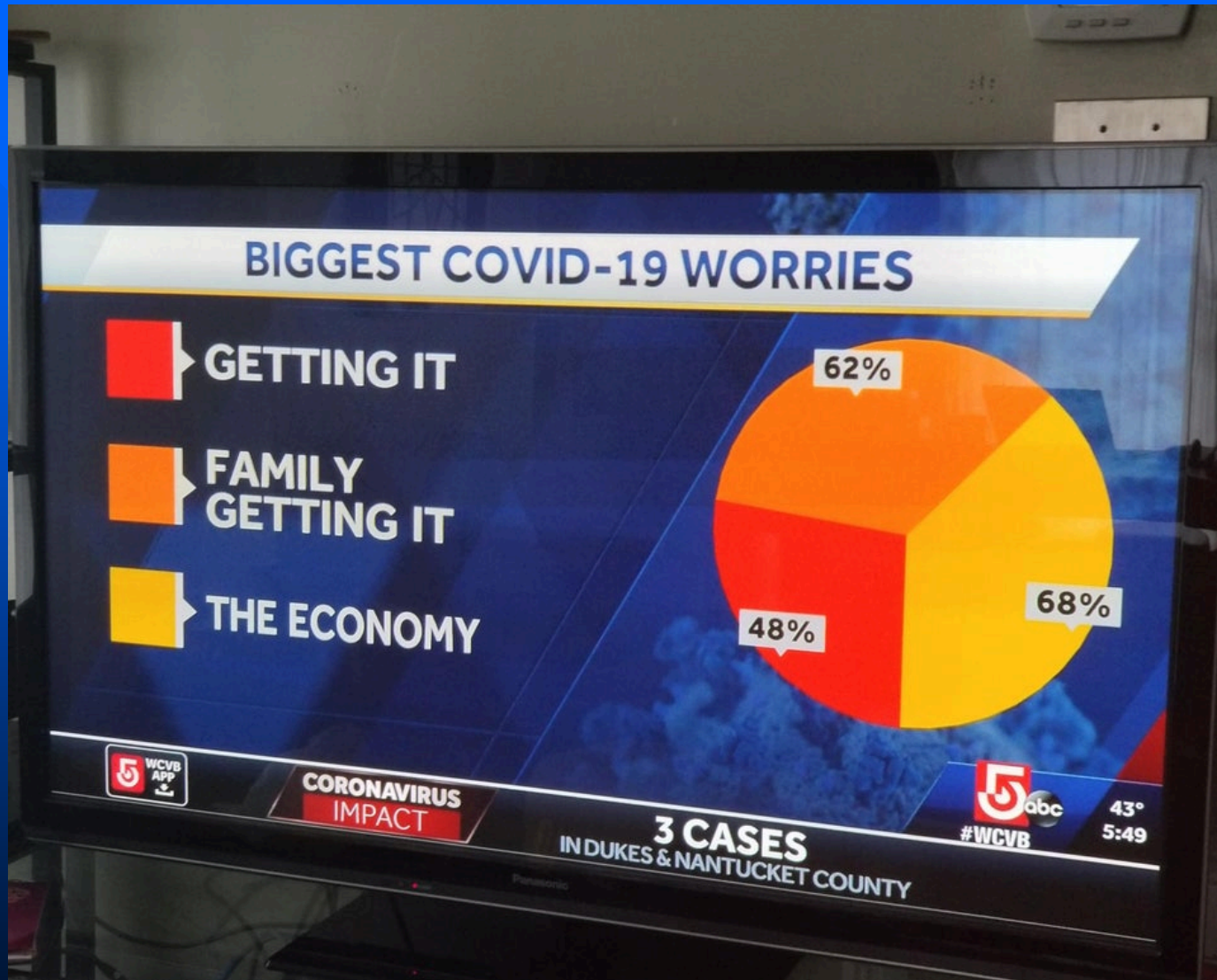
Dynamics of the spread of COVID-19 in Russia.

Over the past 24 hours, 500 people have been infected with coronavirus in 26 regions, 55 have recovered, and nine have died.

The total number of infected is 2337,
121 people recovered, 17 patients
died.



A Pie Chart out of 178% – The Original



WCVB Channel 5 Boston (Graph no longer available from source)

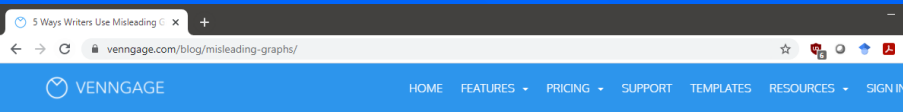
A Pie Chart out of 178% – Bad Graph

Rules Followed

- Rule 1: Show as little data as possible (minimize the data density): Only three possible outcomes are shown.
- Rule 12: If it has been done well in the past, think of a new way to do it: A pie chart is the wrong graph type here; why not a regular bar chart or Cleveland-style dot plot?



A Pie Chart out of 178% – Discussion (1)



5 Ways Writers Use Misleading Graphs To Manipulate You [INFOGRAPHIC]

BY RYAN MCCREADY, APR 17, 2020

3. Featuring the wrong graph

Now that we looked at a few pretty terrible data visualizations, let's see if we can improve some that just used the wrong graphs.

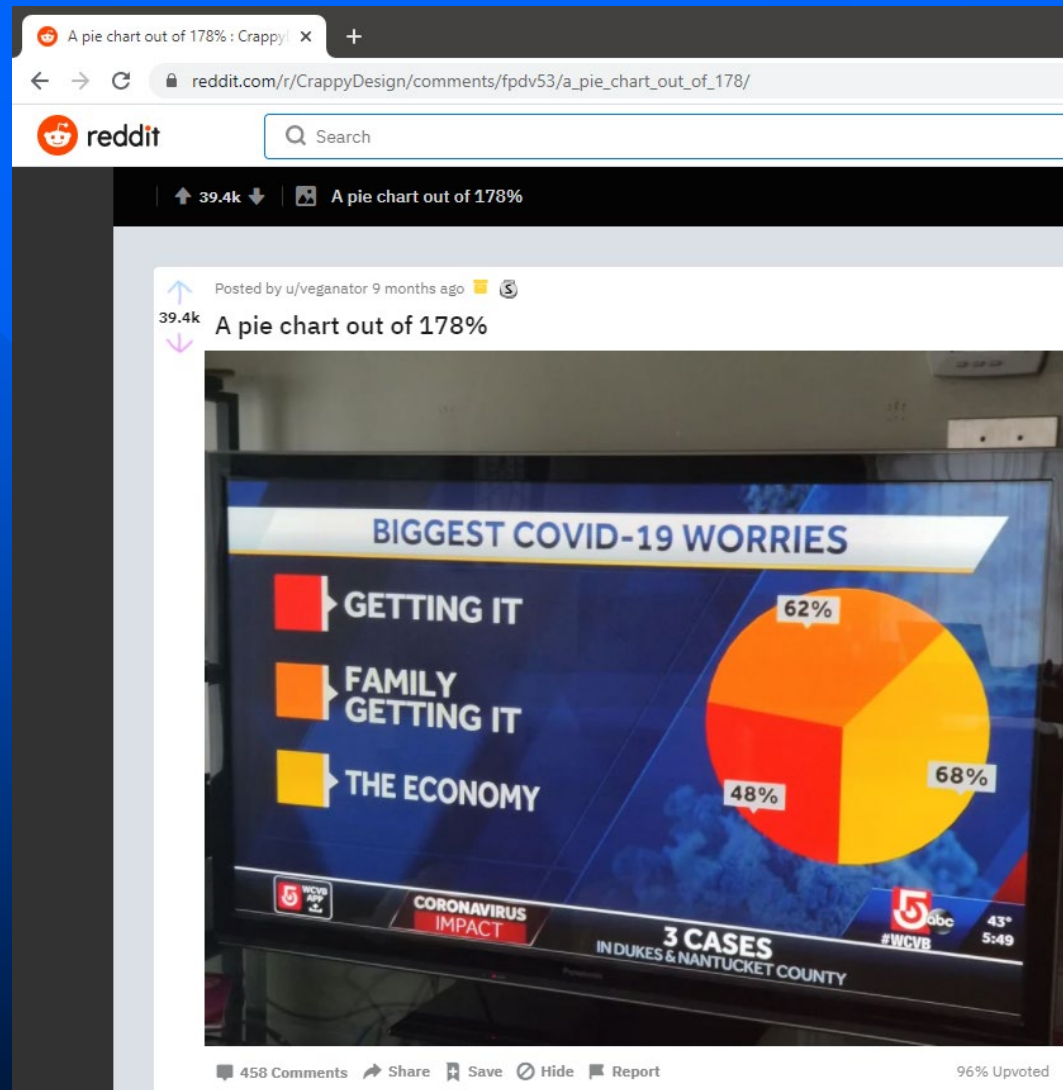
Like I said earlier, using the wrong graph isn't a great excuse but it's better than being intentionally misleading or lazy with your data.

As you can see in this pie chart, the percentages from each section add up to more than 100%. This is a common problem when people try to visualize survey data that has multiple answers.



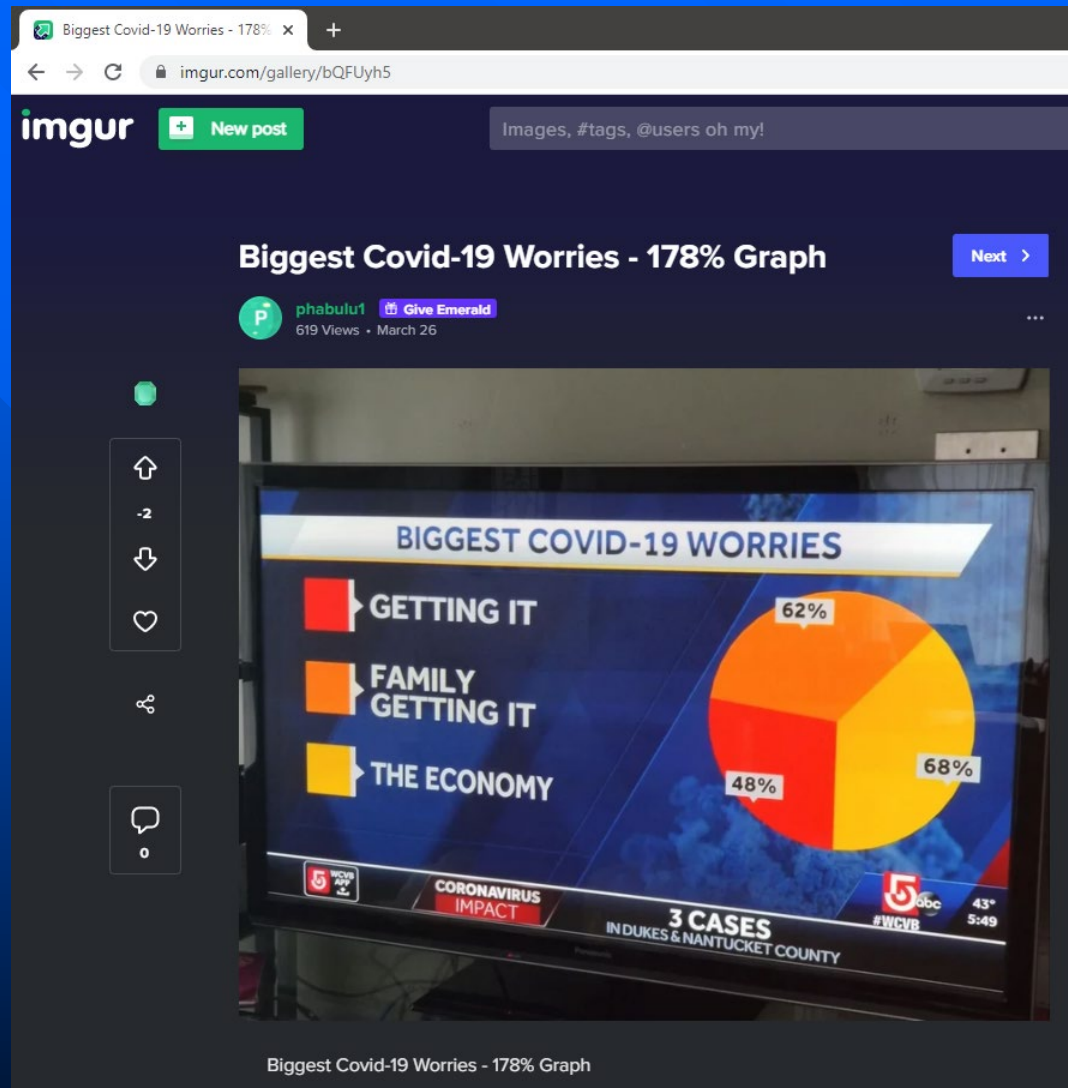
<https://venngage.com/blog/misleading-graphs/>

A Pie Chart out of 178% – Discussion (2)



https://www.reddit.com/r/CrappyDesign/comments/fpdv53/a_pie_chart_out_of_178/

A Pie Chart out of 178% – Discussion (3)



<https://imgur.com/gallery/bQFUyh5>

A Pie Chart out of 178% – Discussion (4)

Isaac Reyes - Data Storyteller - Isaac Reyes - Data Storyteller - Isaac Reyes - Data Storyteller

m.facebook.com/isaac.reyes/photos/a.510013149194230/1344401419088728/

facebook

Isaac Reyes - Data Storyteller is on Facebook. To connect with Isaac Reyes - Data Storyteller, log into Facebook.

Log In

or

Join



Isaac Reyes - Data Storyteller

I'm more worried about this chart than covid19 WCVB Channel 5 Boston

Timeline Photos · March 2020 ·

View Full Size

<https://m.facebook.com/isaac.reyes/photos/a.510013149194230/1344401419088728/>

Covid-19 Cases in the US – The Original

COVID-19 CASES IN THE U.S.



JOHN HOPKINS TRACKER



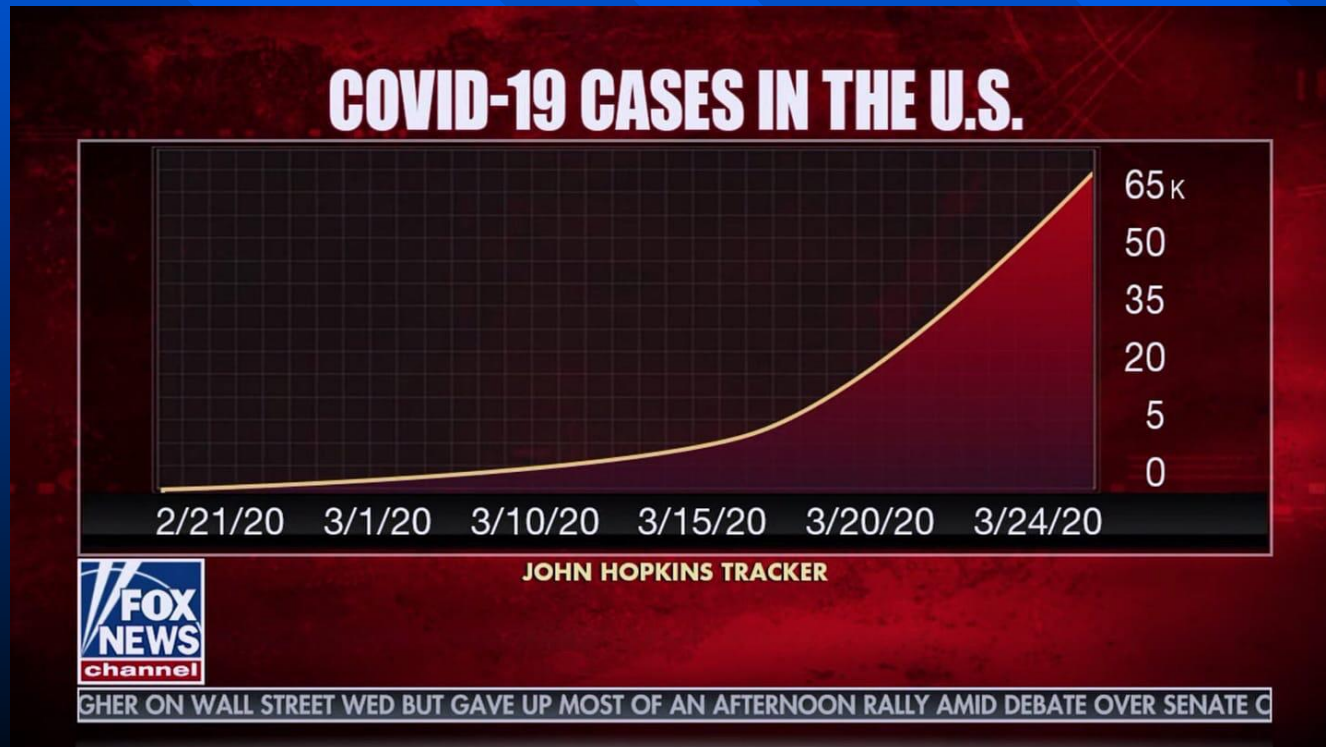
GHIER ON WALL STREET WED BUT GAVE UP MOST OF AN AFTERNOON RALLY AMID DEBATE OVER SENATE C

Fox News Channel (Graph no longer available from source)

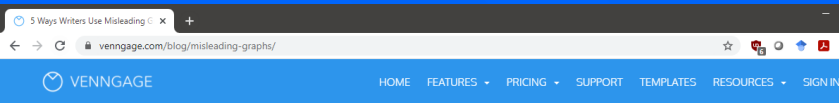
Covid-19 Cases in the US – Bad Graph

Rules Followed

- Rule 6: Change scales in mid-axis: Horizontal axis labels are placed after 9, 9, 5, 5, and 4 days.
- Rule 6: Change scales in mid-axis: Vertical axis labels are placed after 5K and then after every 15K.



Covid-19 Cases in the US – Discussion (1)

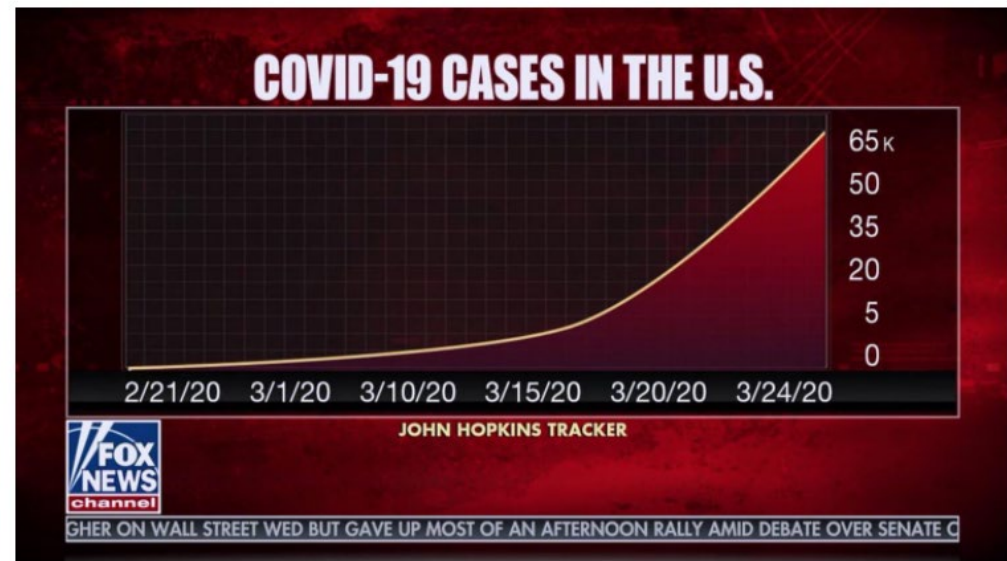


5 Ways Writers Use Misleading Graphs To Manipulate You [INFOGRAPHIC]

BY RYAN MCCREADY, APR 17, 2020

4. Manipulating the Y & X-axis

Let's get back to some actually misleading graphs, starting with this example from Fox News:

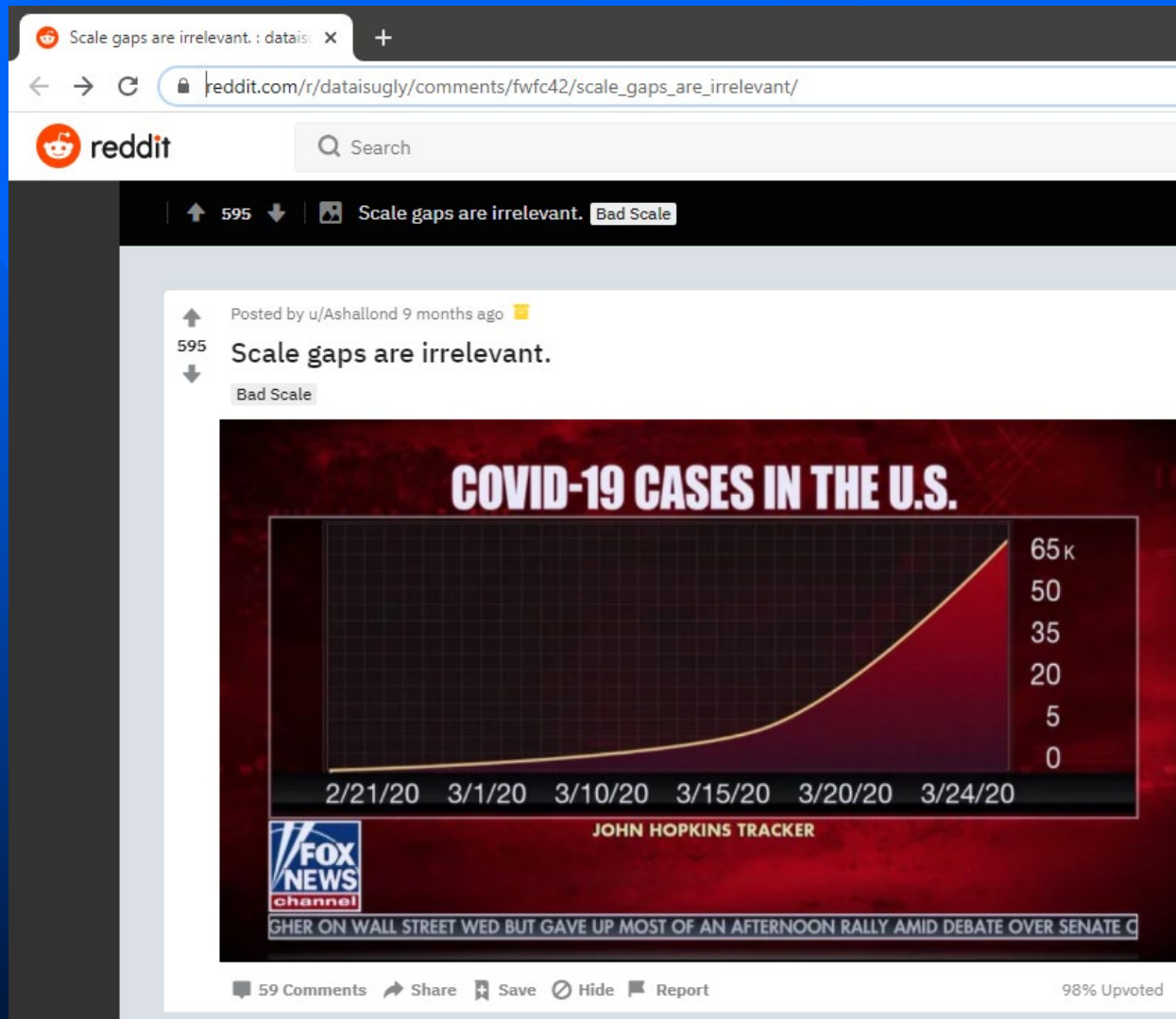


[Source](#)

As you can see, both the x-axis and y-axis have been manipulated, which is very common on graphs like this.

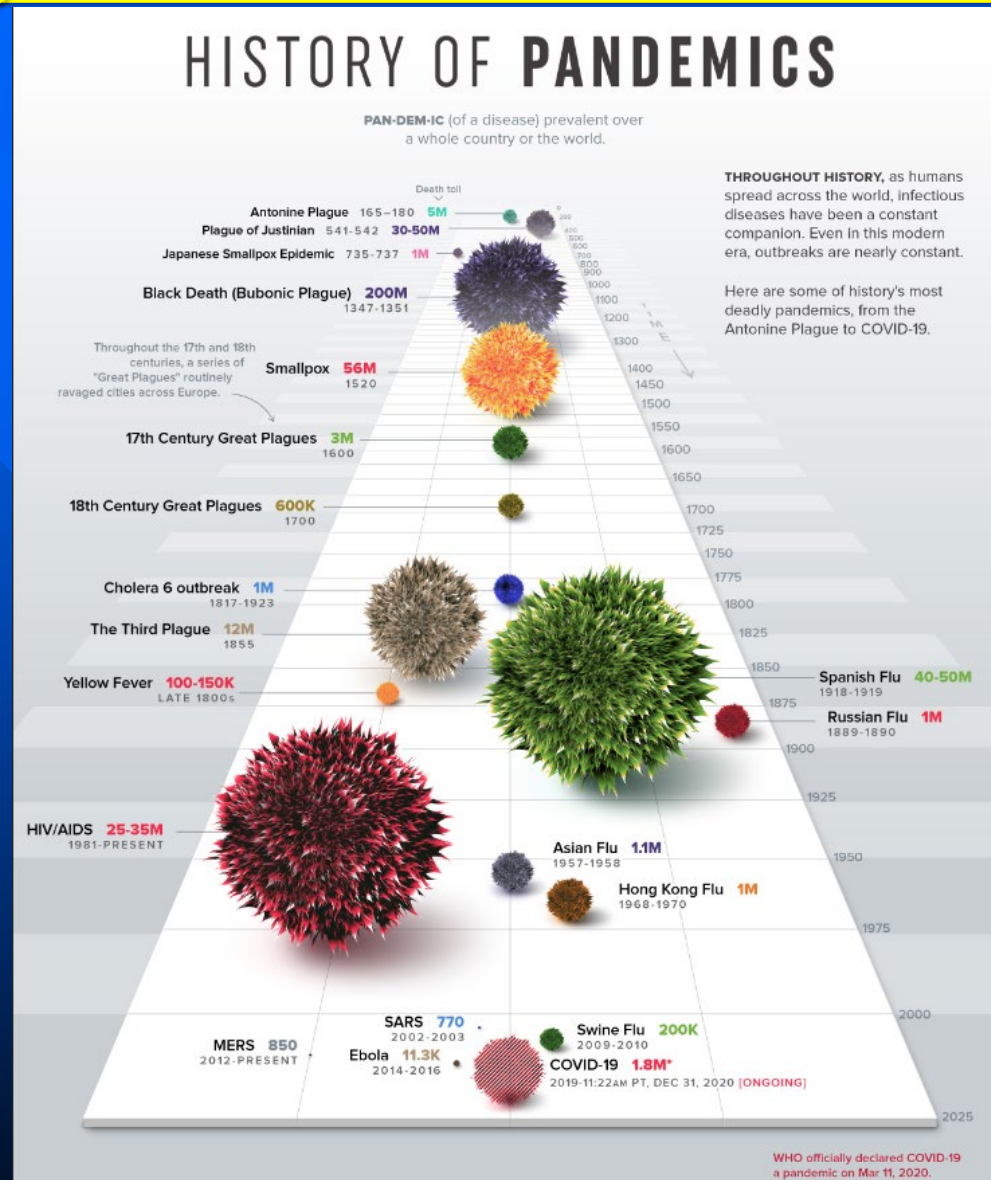
<https://venngage.com/blog/misleading-graphs/>

Covid-19 Cases in the US – Discussion (2)



https://www.reddit.com/r/dataisugly/comments/fwfc42/scale_gaps_are_irrelevant/

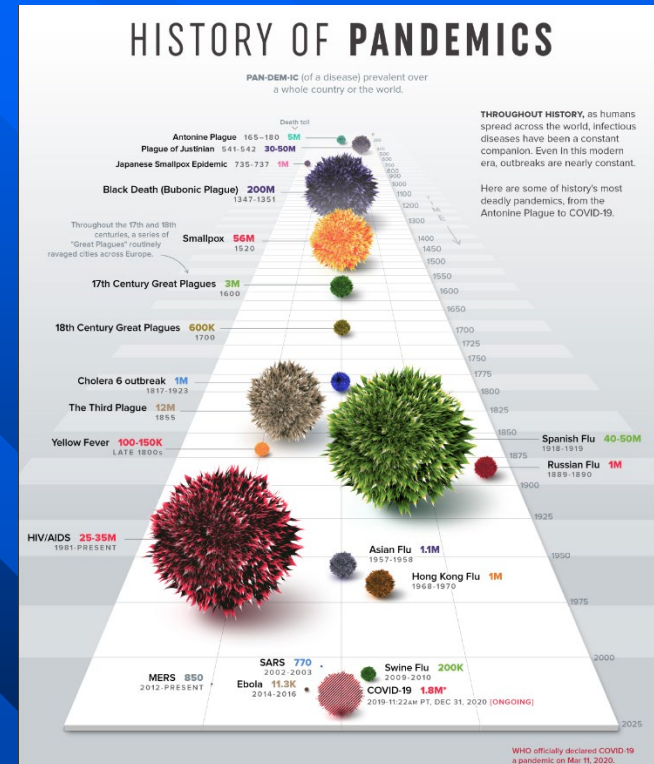
History of Pandemics – The Original



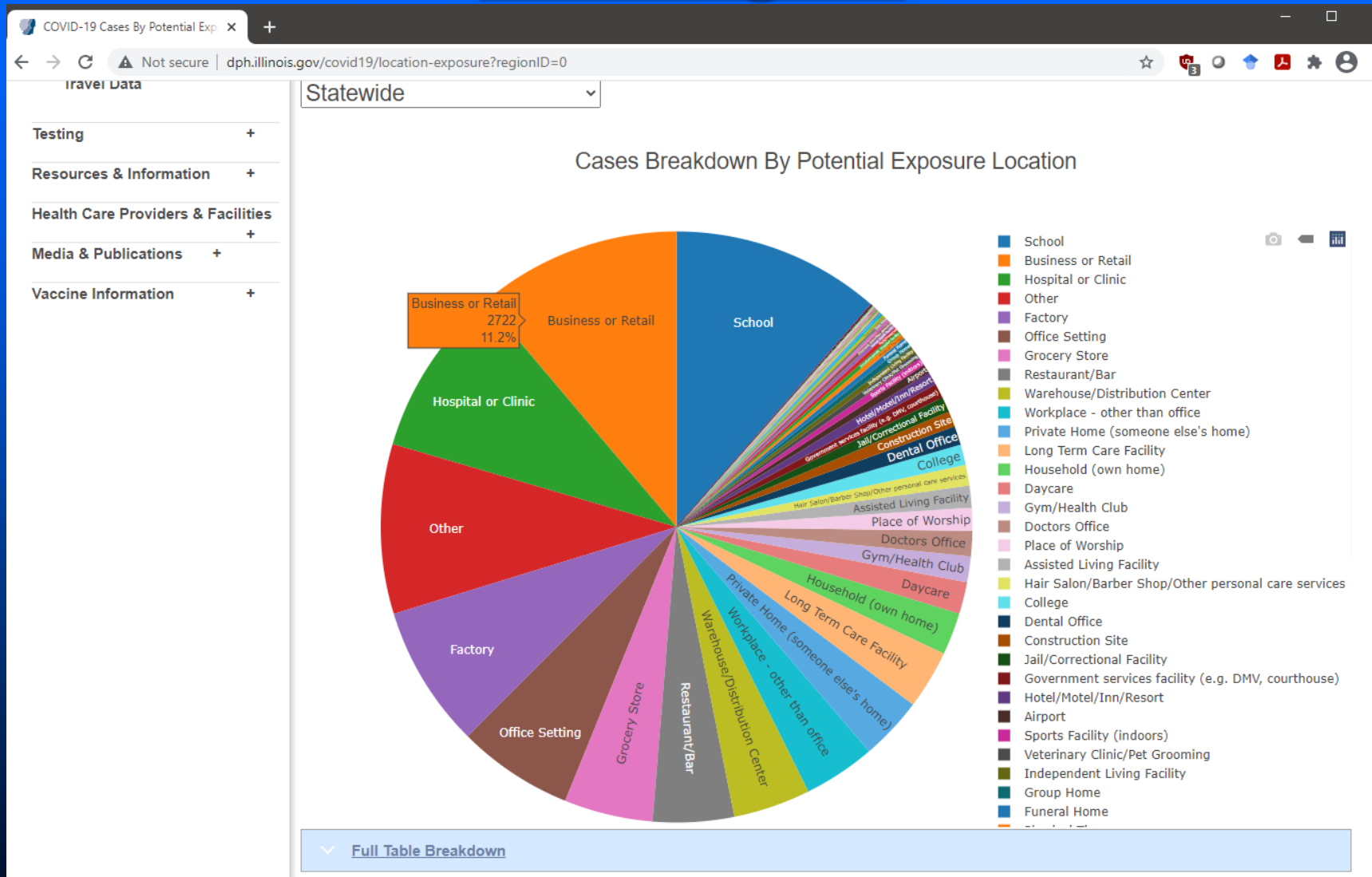
History of Pandemics – Bad Graphs

Rules Followed

- Rule 3: Ignore the visual metaphor altogether: Size of viruses not related to number of deaths.
- Rule 6: Change scales in mid-axis: Vertical time axis is condensed, due to the 3D-perspective view.
- Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously: “Death toll” label almost invisible.
- Rule 11: More is murkier: (a) more decimal places and (b) more dimensions: 3D representation makes some of the viruses/numbers look even bigger.



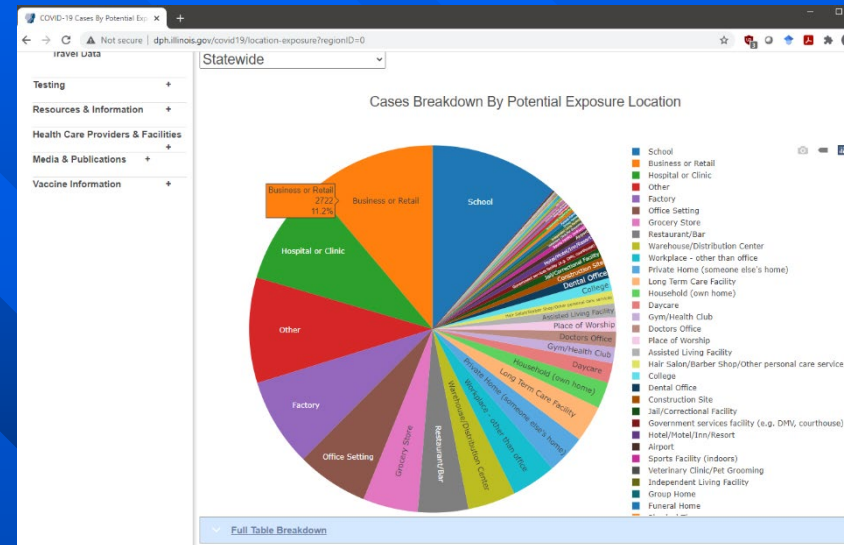
Cases by Potential Exposure Location – The Original



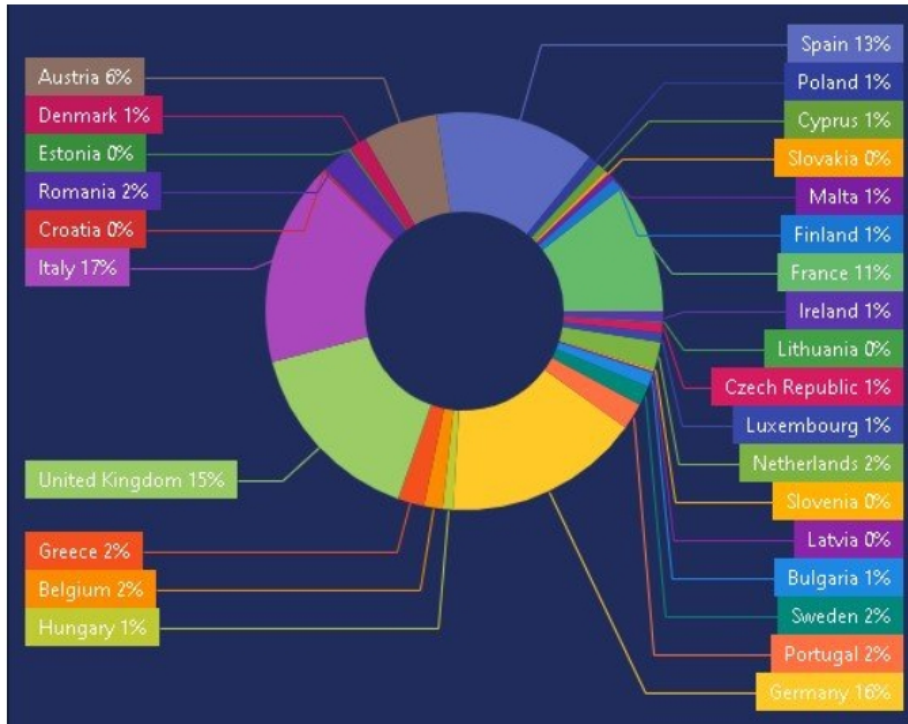
<http://www.dph.illinois.gov/covid19/location-exposure?regionID=0>

Cases by Potential Exposure Location – Bad Graph Rules Followed

- Rule 2: Hide what data you do show (minimize the data/ink ratio): Some of the categories cannot be identified at all.
- Rule 9: Alabama first!: Segments in the pie chart are not arranged in any meaningful order at all.
- Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously: Many of the labels are too small to read.



Media Coverage in Europe – The Original



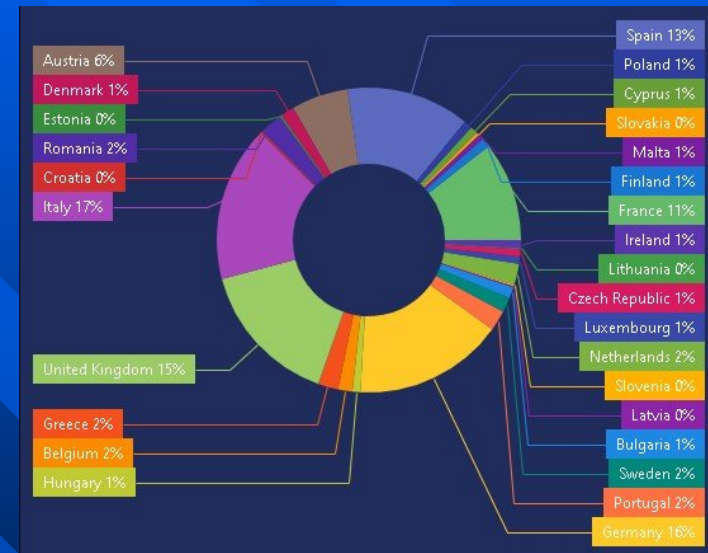
Pie graph shows us the percentage of **European countries** in relation to the Coronavirus. How much were these countries mentioned with the COVID-19 in **the media**?

Top five countries:

Italy 17%
Germany 16%
England 15%
Spain 13%
France 11%

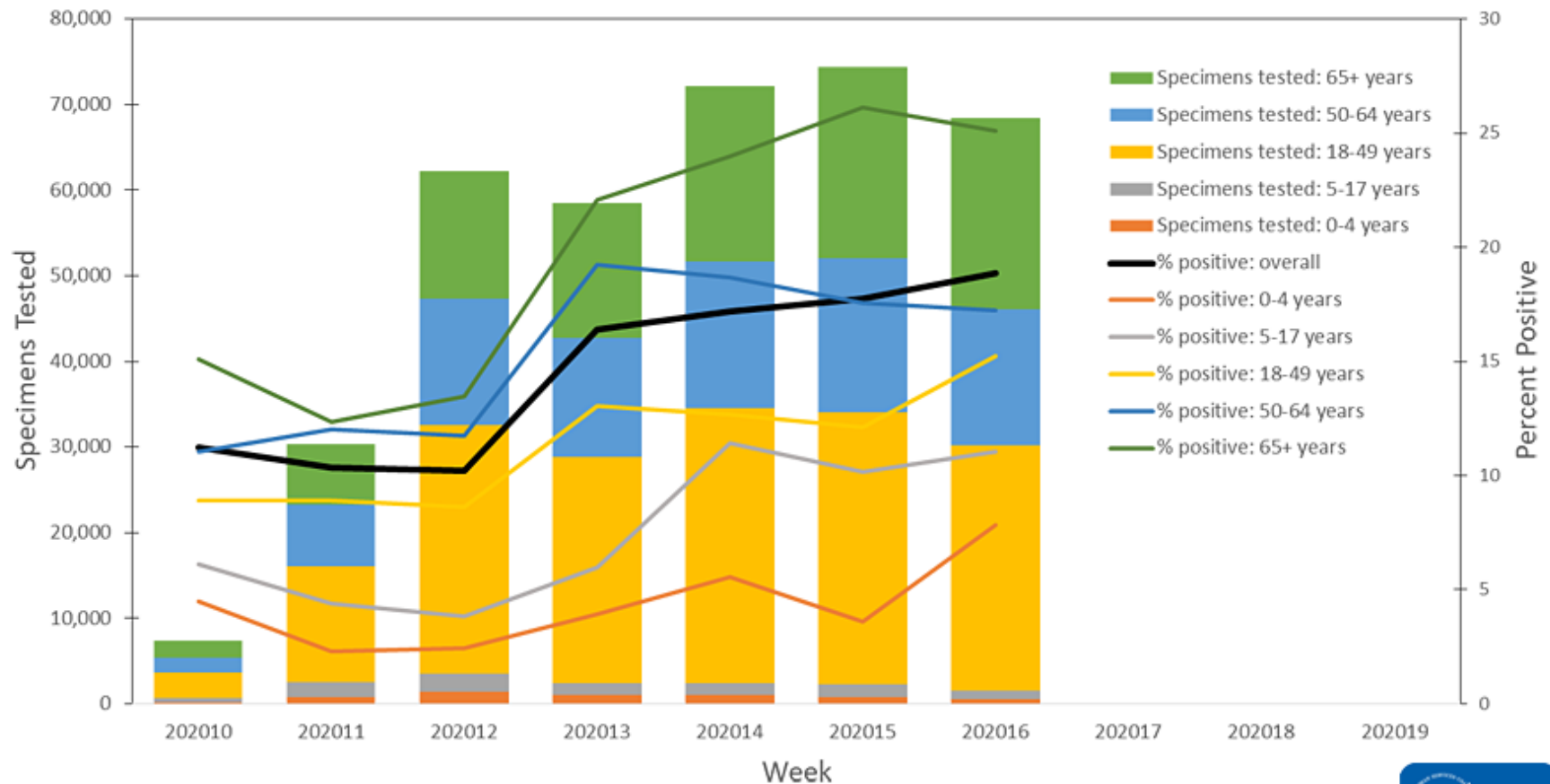
Media Coverage in Europe – Bad Graph Rules Followed

- Rule 9: Alabama first!: Segments in the donut chart are not arranged in any meaningful order at all.
- Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously: Russia and Switzerland (and others) are not shown; these seem to be countries of the “European Union” (and not “European Countries”).
- Rule 12: If it has been done well in the past, think of a new way to do it: Why showing 0% areas (i.e., countries) in the donut chart?



CDC Public Health Labs – The Original

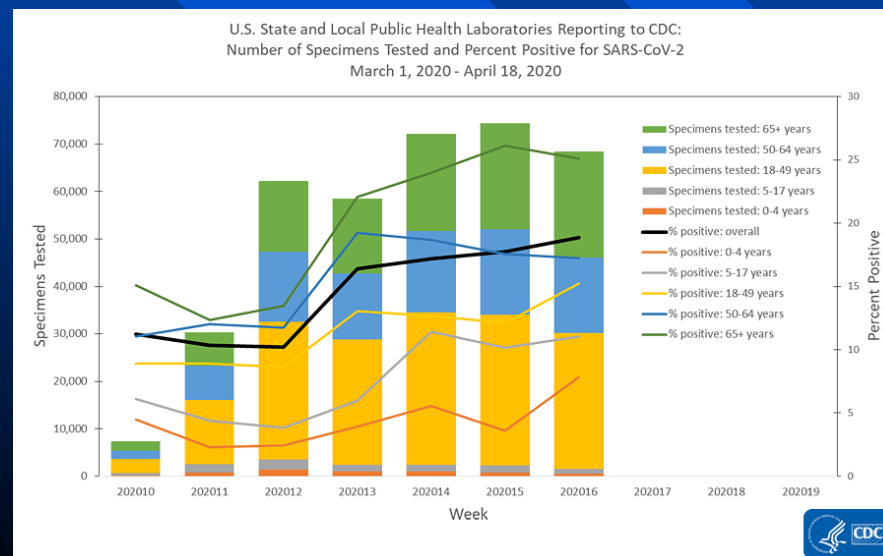
U.S. State and Local Public Health Laboratories Reporting to CDC:
Number of Specimens Tested and Percent Positive for SARS-CoV-2
March 1, 2020 - April 18, 2020



CDC Public Health Labs – Bad Graph

Rules Followed

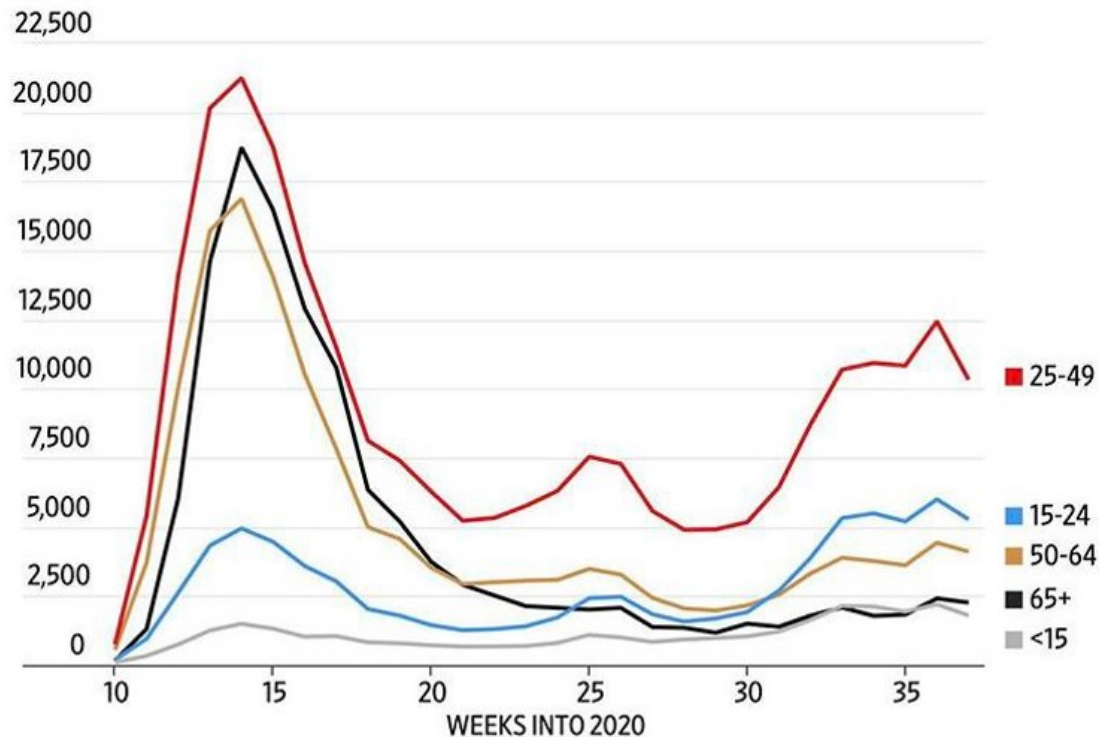
- Rule 8: Jiggle the baseline: The uneven (jiggled) base line makes it difficult to see trends for the 65+ (green), 50-64 (blue), and 18-49 (yellow) age groups.
- Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously: Instead of using labels such as “202010”, separate month and week and label as “#10/2020” or “2020-10” for example.



Younger People Surge – The Original

Younger People Are Driving the Current Surge in Covid-19 Cases Across Europe

Confirmed Covid-19 cases in 17 EU countries by age group



Note: Countries include Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, Germany, Iceland, Ireland, Latvia, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal and Sweden

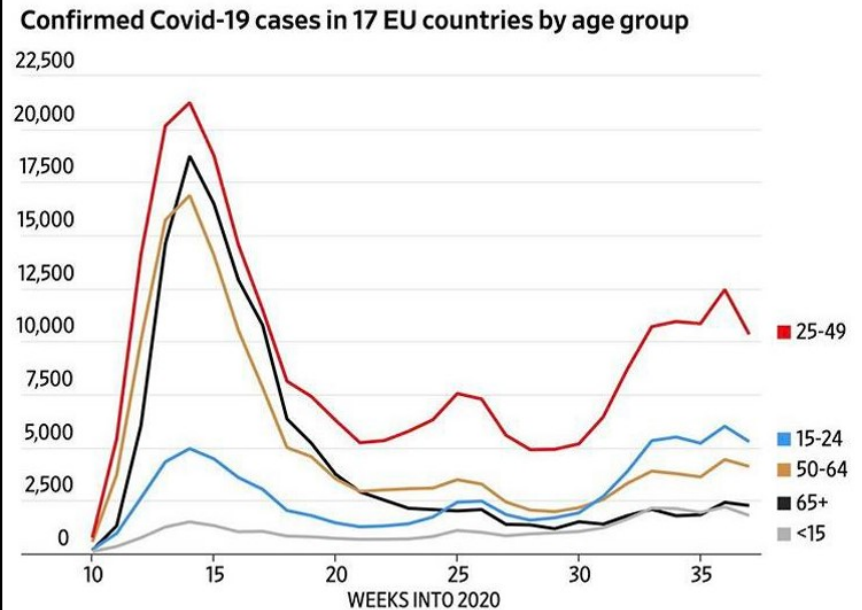
Source: European Surveillance System, part of the European Center for Disease Prevention and Control

Younger People Surge – Bad Graph

Rules Followed

- Rule 7: Emphasize the trivial (ignore the important): The 25-49 year category spans 25 years; no surprise that this contains more cases than the age groups that only span 10 or 15 years.

Younger People Are Driving the Current Surge in Covid-19 Cases Across Europe



Note: Countries include Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, Germany, Iceland, Ireland, Latvia, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal and Sweden

Source: European Surveillance System, part of the European Center for Disease Prevention and Control

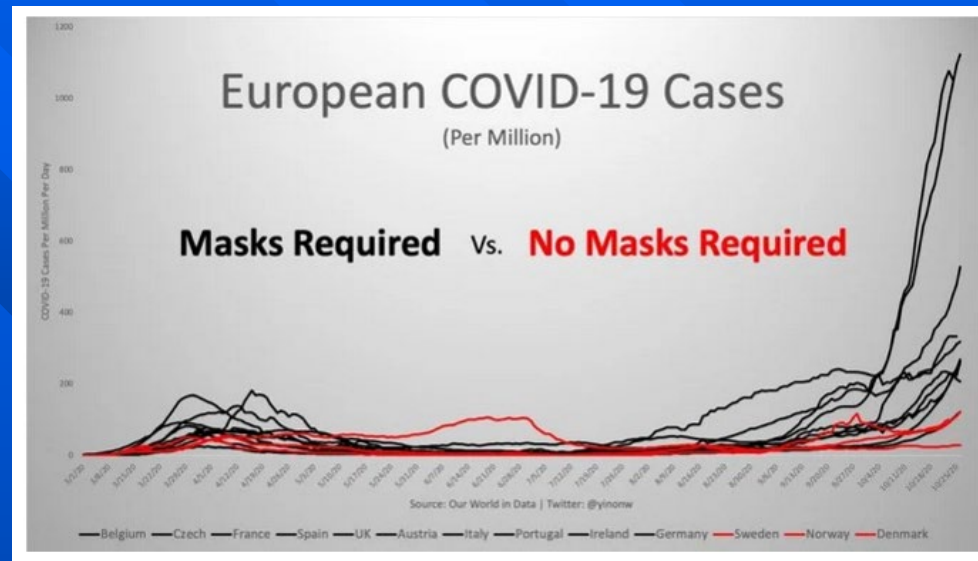
European Comparison – The Original



European Comparison – Bad Graph

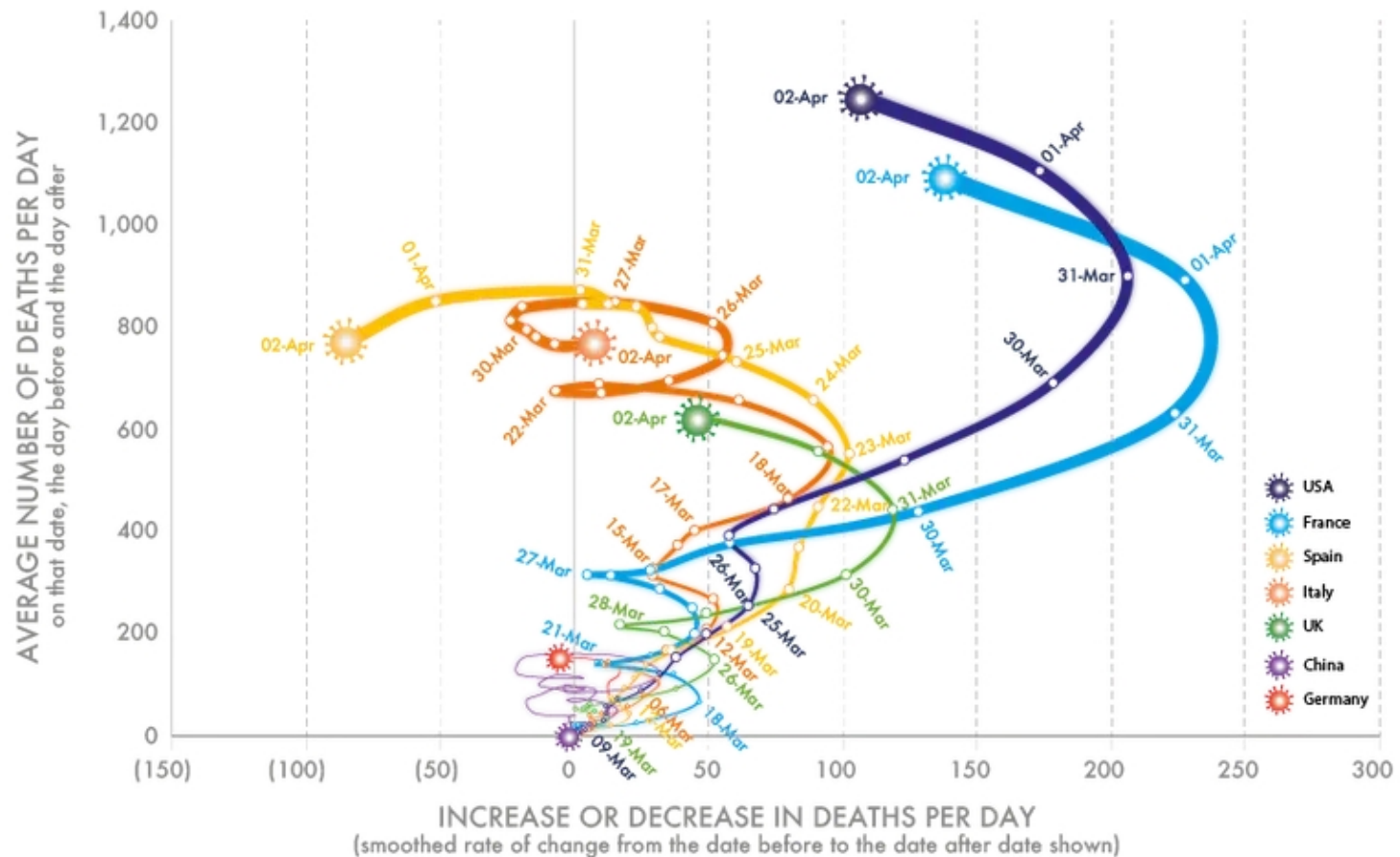
Rules Followed

- Rule 5: Graph data out of context: Why these countries, but omitting the Netherlands, Russia, and Switzerland for example?
- Rule 5: Graph data out of context: No further context information on lockdowns, school & business closures, contact tracing, etc. provided.
- Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously:
Impossible to identify which line belongs to which country.



Mortality due to Covid-19 – The Original (1)

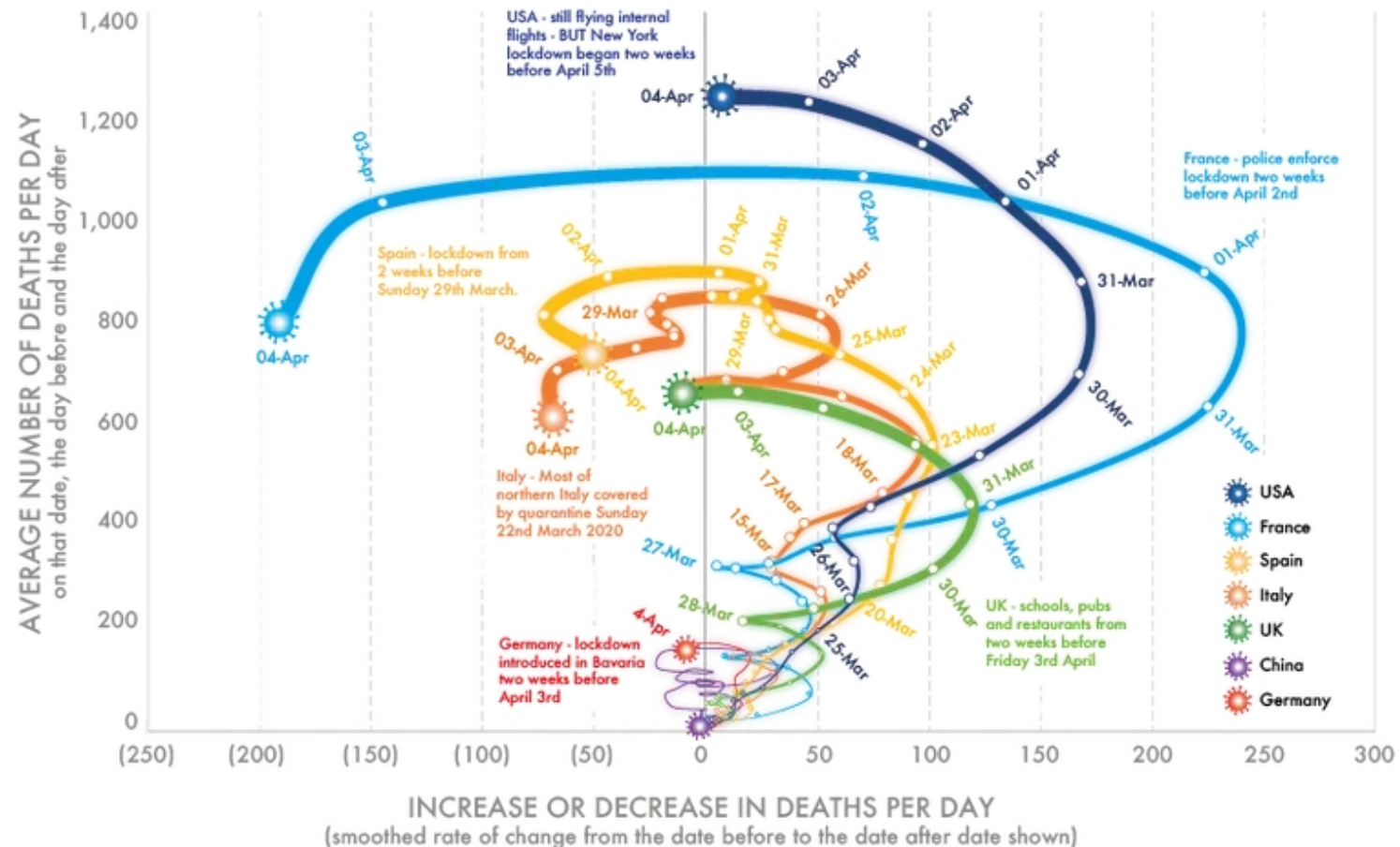
Mortality in seven countries attributed to COVID-19 (January 23 to April 2, 2020).



Danny Dorling/Kirsten McClure, Author provided

Mortality due to Covid-19 – The Original (2)

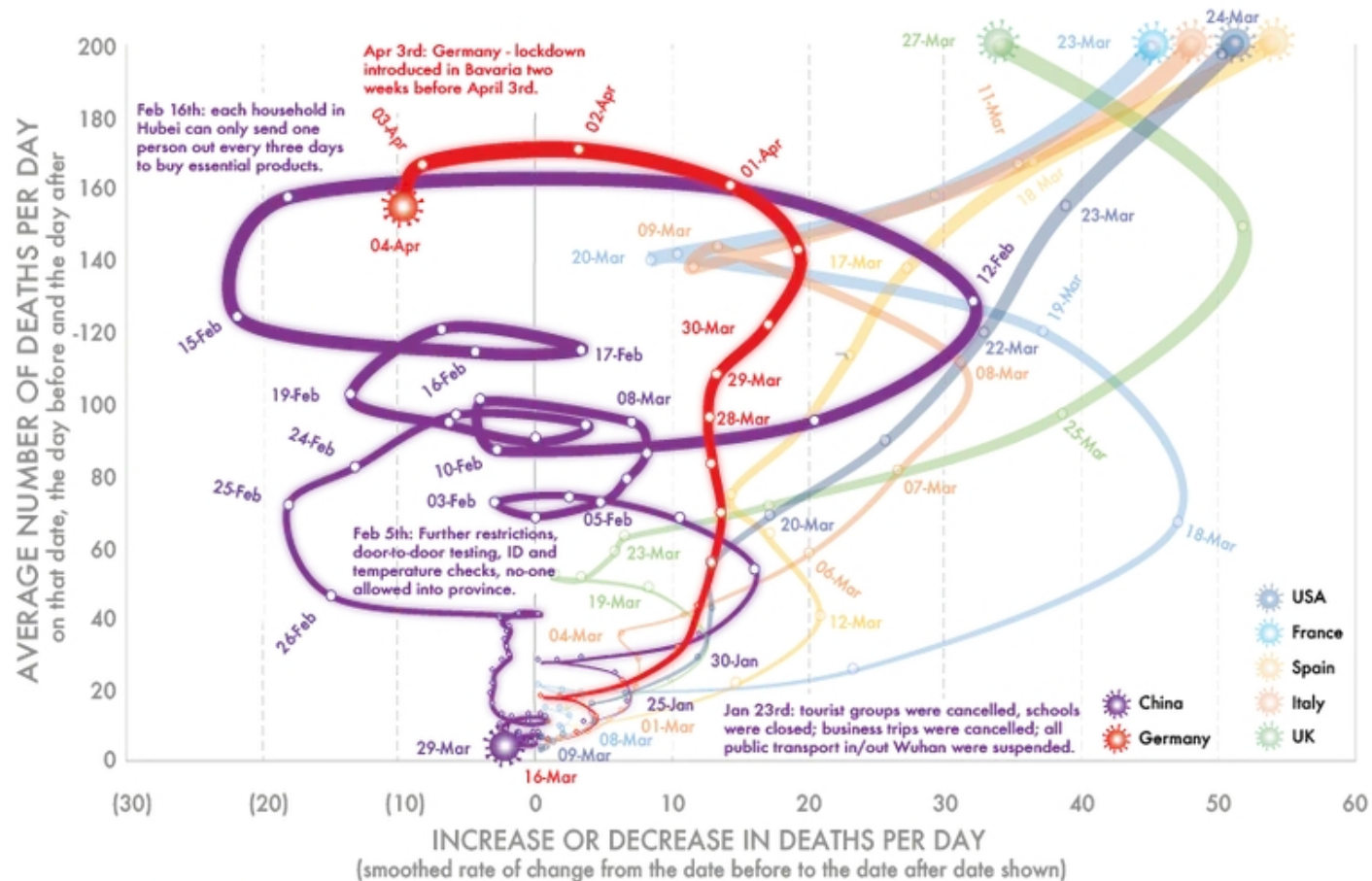
Mortality in seven countries attributed to COVID-19 (January 23 to April 4, 2020).



Danny Dorling/Kirsten McClure, Author provided

Mortality due to Covid-19 – The Original (3)

Mortality in China and Germany attributed to COVID-19 (January 23 to April 4, 2020)

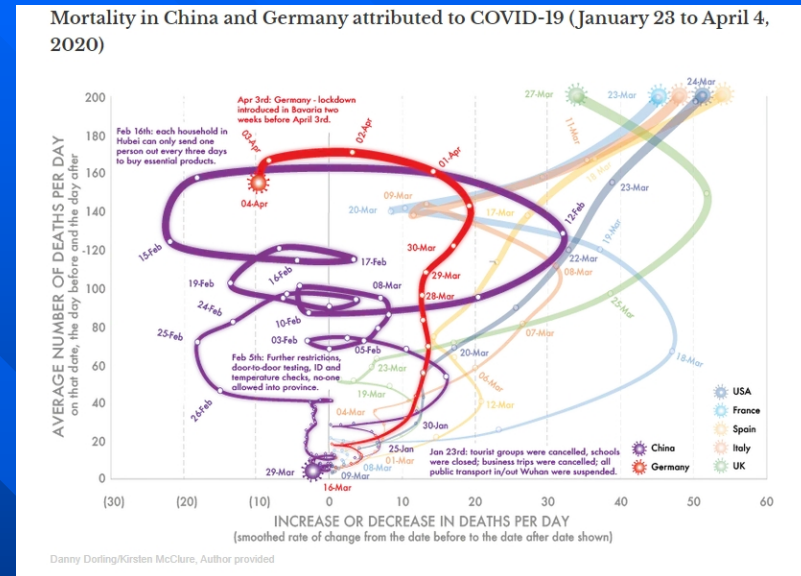


Danny Dorling/Kirsten McClure, Author provided

Mortality due to Covid-19 – Bad Graph

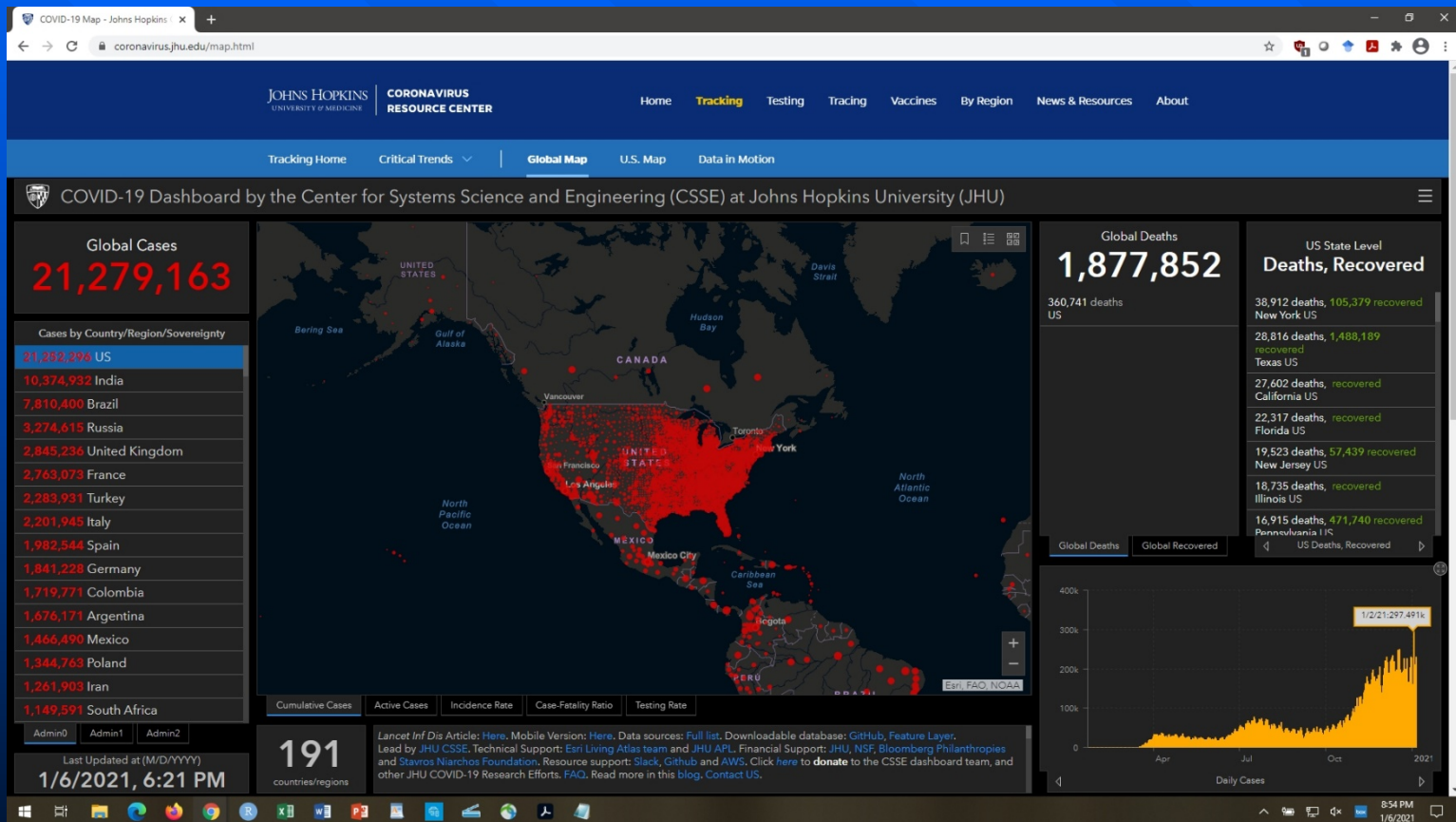
Rules Followed

- Rule 2: Hide what data you do show (minimize the data/ink ratio): There is a lot of overplotting in the lower part of the graph
- Rule 3: Ignore the visual metaphor altogether: Why changing the line thickness? Or does this represent a fourth variable?
- Rule 5: Graph data out of context: Why these seven countries, but leaving out others with high numbers of cases in March (e.g., the Netherlands, Belgium, Iran)?
- Rule 10: Label: (a) illegibly, (b) incompletely, (c) incorrectly, and (d) ambiguously: Many labels difficult to read.
- Rule 12: If it has been done well in the past, think of a new way to do it: The graph tries to display three (or four?) variables (date, increase/decrease per day, average number of deaths per day) in a two-dimensional layout.

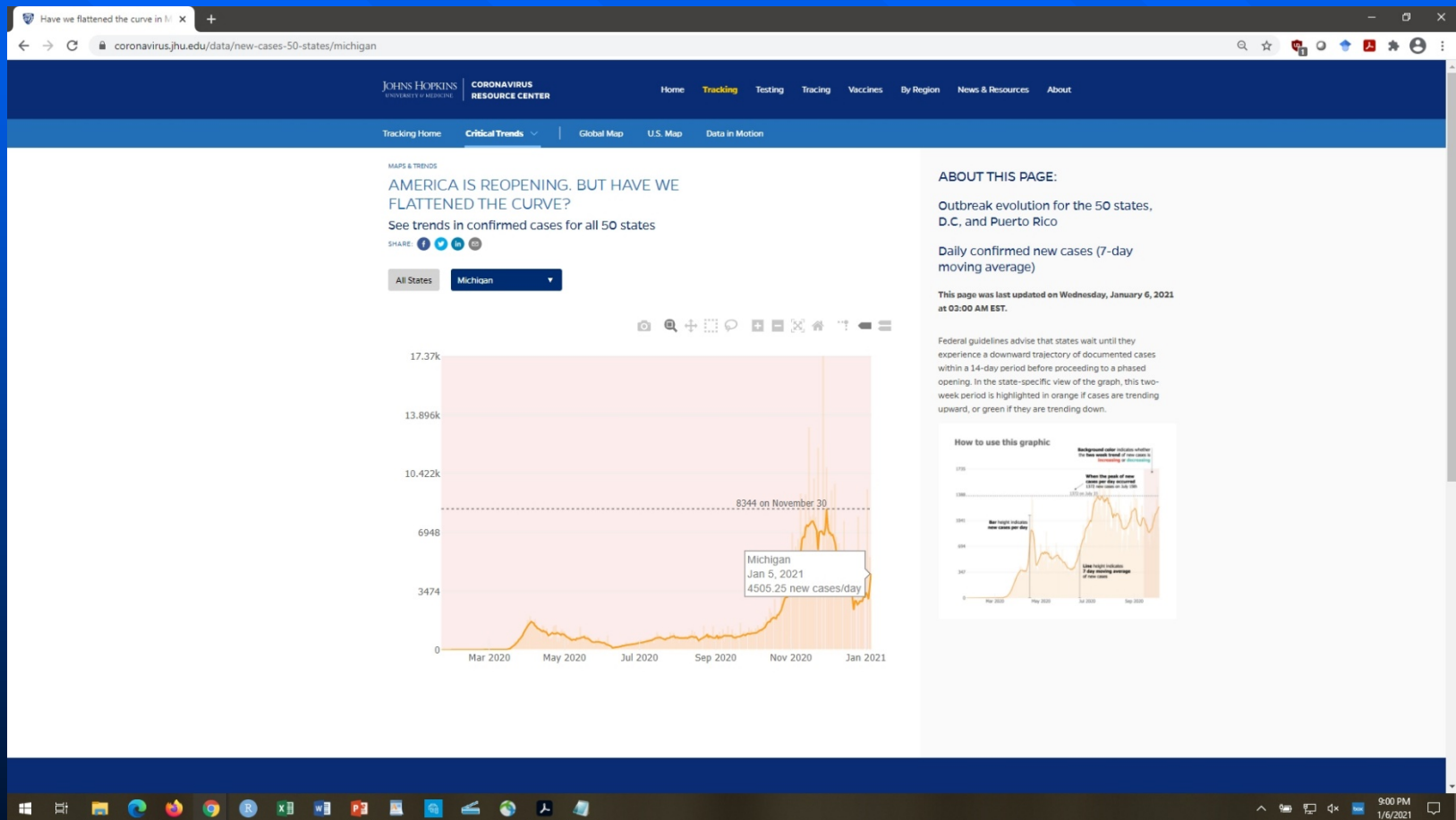


The Good Ones

<https://coronavirus.jhu.edu/map.html>

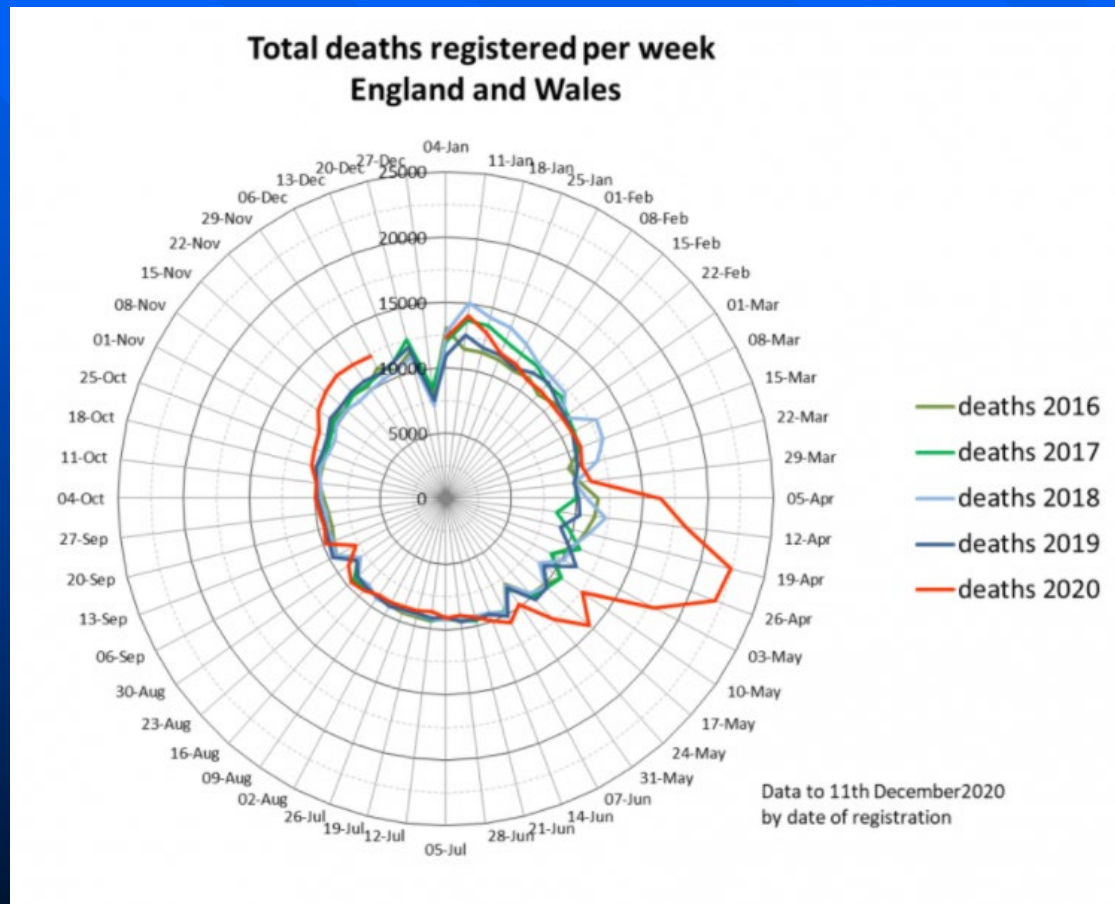


Johns Hopkins University (JHU): Maps & Trends – America is Reopening. But have We Flattened the Curve?

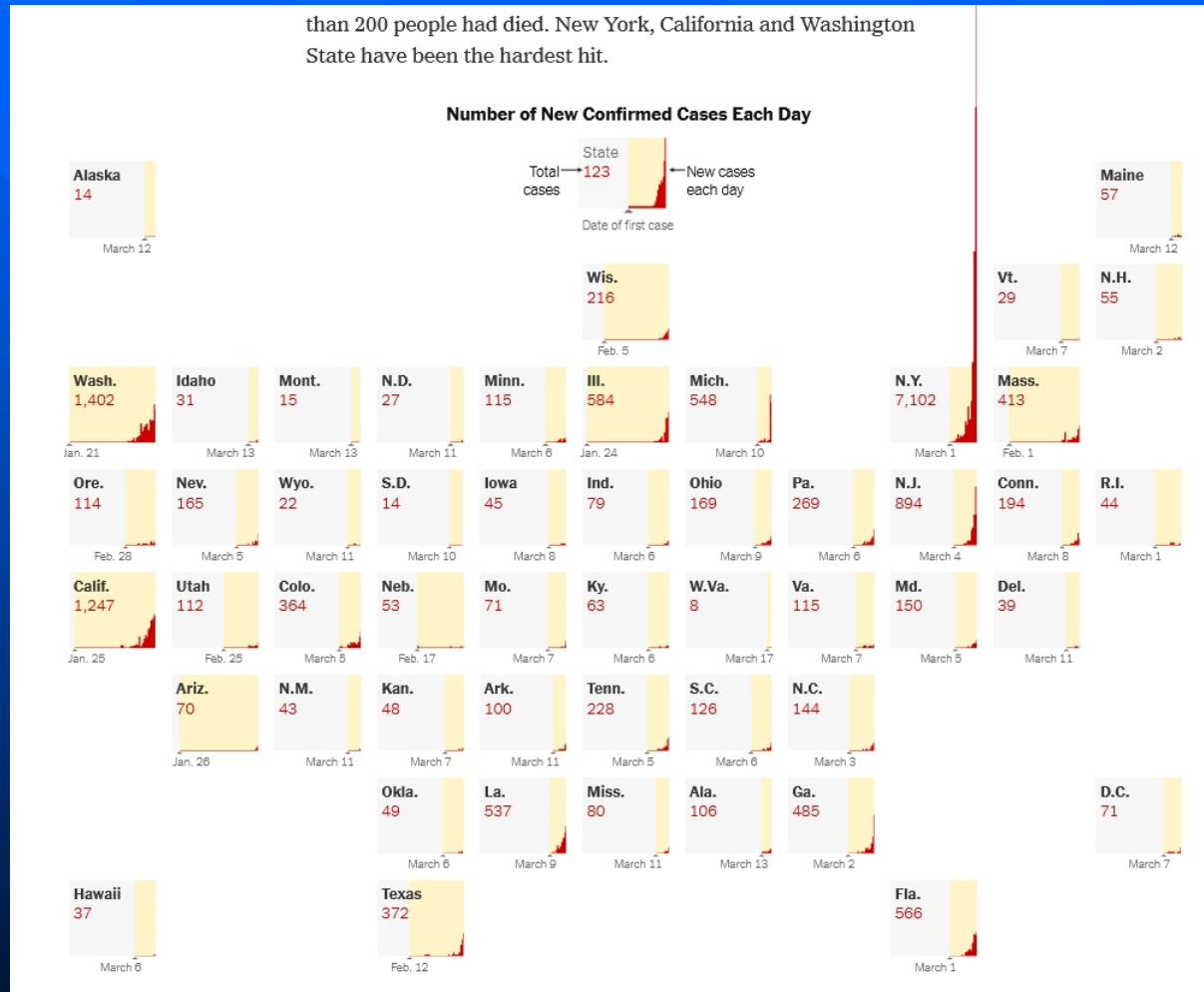


<https://coronavirus.jhu.edu/data/new-cases-50-states/michigan>

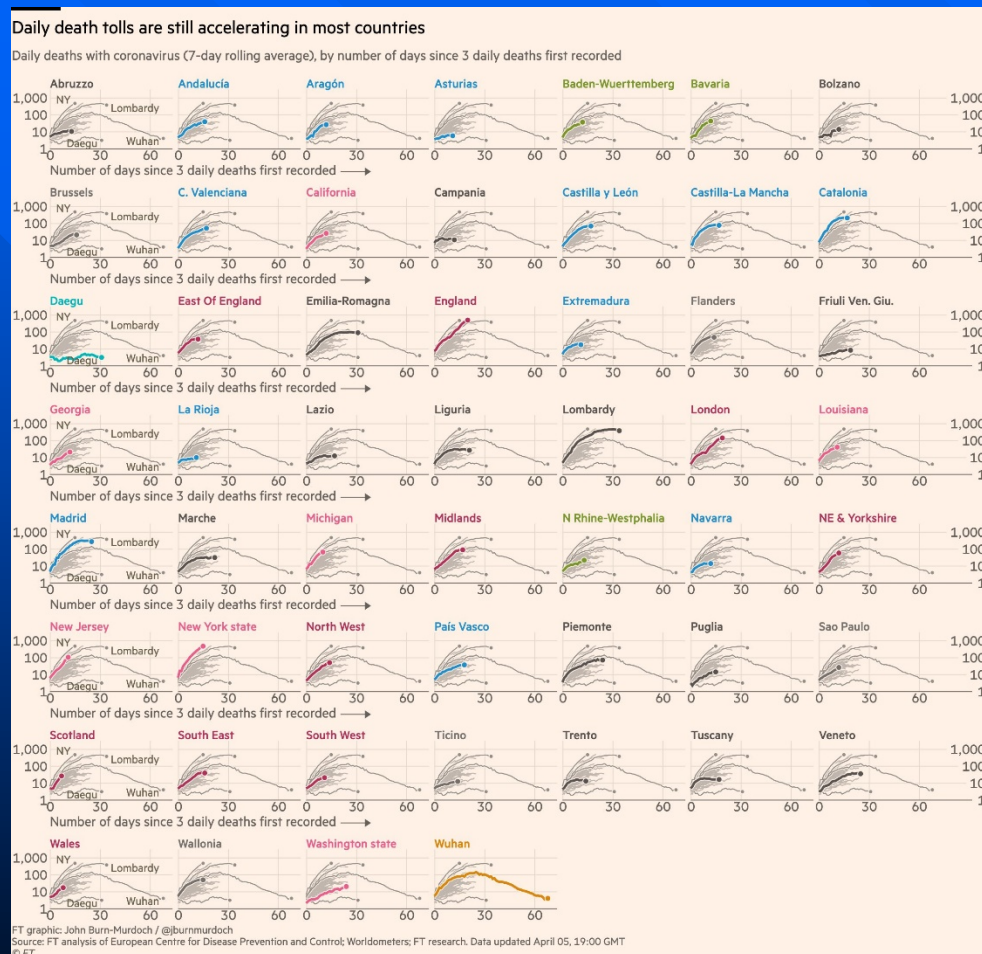
The Centre for Evidence-Based Medicine: COVID-19 – Florence Nightingale Diagrams of Deaths in England & Wales



The New York Times: Watch How the Coronavirus Spread Across the United States

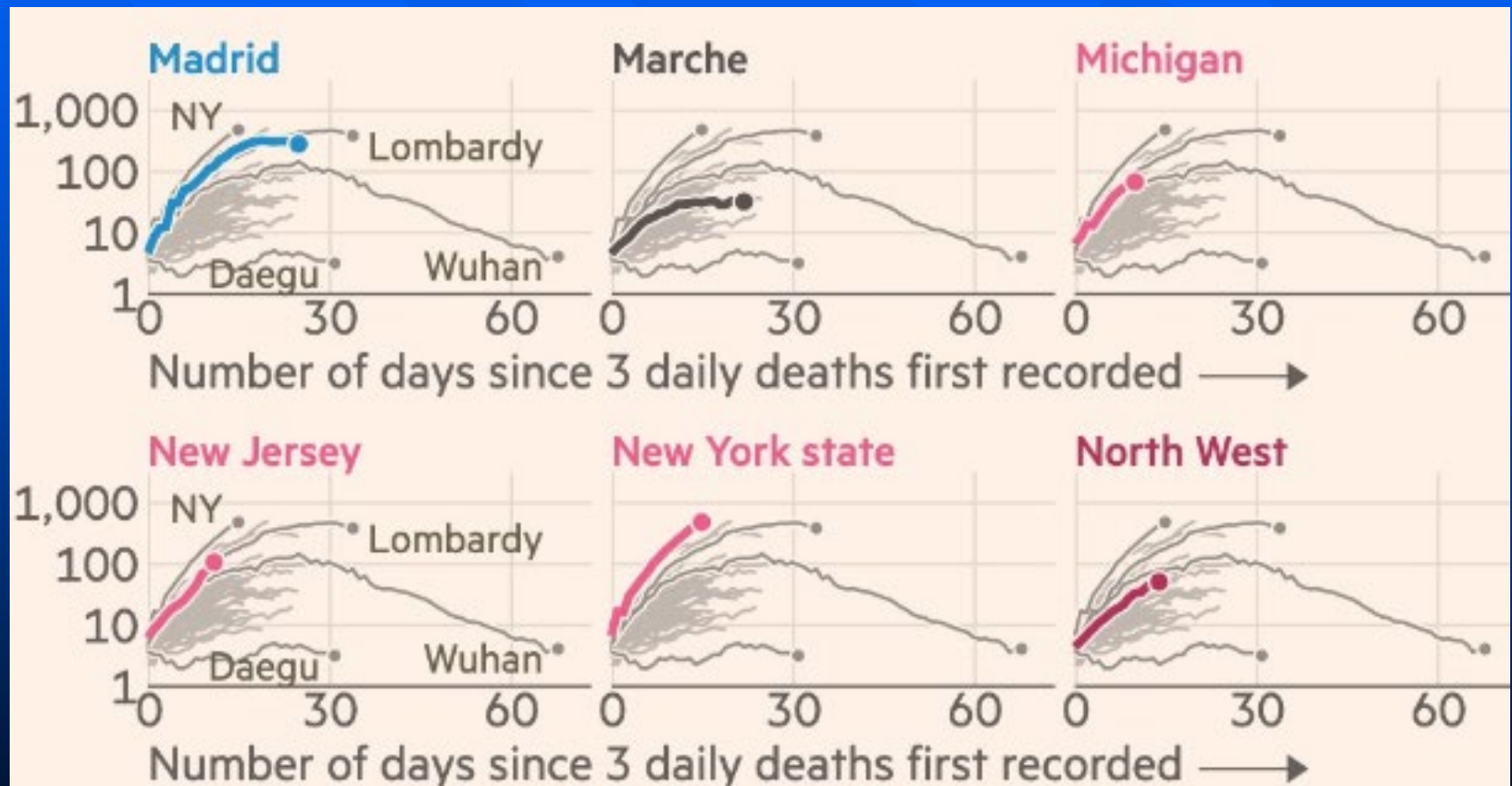


John Burn-Murdoch: Small Multiples of Daily Deaths in Subnational Regions



<https://twitter.com/jburnmurdoch/status/1246904821170483201>

John Burn-Murdoch: Small Multiples of Daily Deaths in Subnational Regions - Close-Up



Summary & Discussion

Possible Explanations for Misleading Graphs

- Graphs should **intentionally** mislead!
- **Lack of Knowledge #1:**
Graphs constructed by non-statisticians, e.g., graphic/web designers.
- **Lack of Knowledge #2:**
Even teachers of statistics often do not know what the correct graphs should look like.

Conclusion ???

Sales data for the City of S X

← → ↺ sanmateorealestateblog.com/2007/10/08/never-believe-any-statistics-you-havent-falsified-yourself/ ☆ 🔒 📄 📱 ☰

Sales data for the City of San Carlos based on semi-annual data from 1998 to 2007

OCTOBER 8, 2007 BY VICKI MOORE

LEAVE A COMMENT

Having been in real estate 5 or so years, I never had any use for statistics. Put the house on the market, a week later it sold.

Now it's another story. Statistics are a crucial part of pricing homes for sale and for determining an appropriate offer price for a buyer. There's a constant discussion happening about what the market is doing and where it's doing it; statistics play a vital role in that discussion.

My client has this great quote on his website, ***"Never believe any statistics you haven't falsified yourself."*** Just so you know, I didn't do anything to these statistics to manipulate them. It's a straight pull of information from the Multiple Listing Service for all neighborhoods within the City of San Carlos based on semi-annual data from 1998 to 2007.

Bad Graphs Collections on the Web

Hungry for More Bad Graphs?!?



John Burn-Murdoch ✓ @jburnmurdoch · Apr 3, 2020

NEW: in the @10DowningStreet press conference, a slide showed Britons adhering to lockdown by staying home.

I've got the same @Google source data, and people in UK & US are still out and about much more than elsewhere...

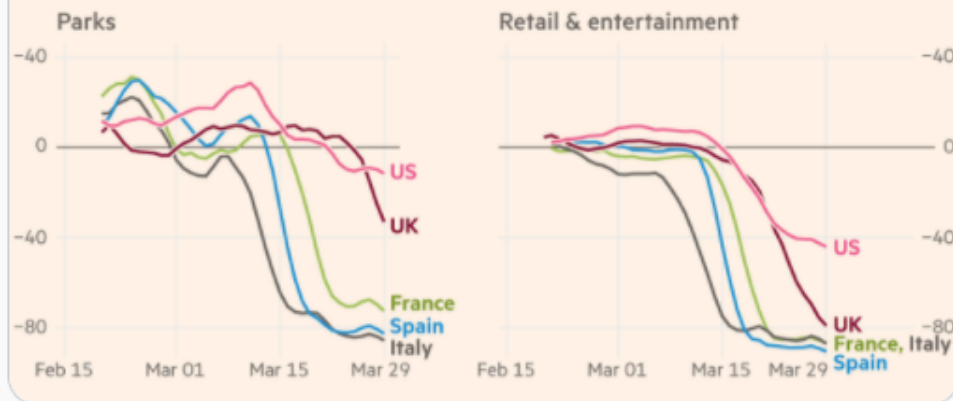
(and no, I'm not a cop 🚔)

More: [ft.com/content/ba6077...](https://www.ft.com/content/ba6077...)

[Show this thread](#)

People in the UK and US are still going out more than those in locked-down European countries

Google mobility index, change from baseline level



12

141

234



<https://twitter.com/jburnmurdoch/status/1246904821170483201>

Hungry for More Bad Graphs?!? – Bon Appetit!

John Burn-Murdoch @jburnmurdoch · Apr 3, 2020

NEW: in the @10DowningStreet press conference, a slide showed Britons adhering to lockdown by staying home.

I've got the same @Google source data, and people in UK & US are still out and about much more than elsewhere...

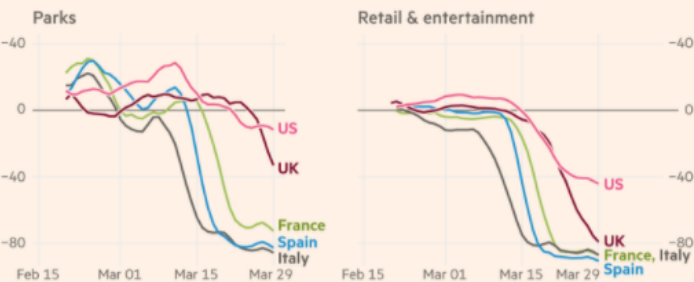
(and no, I'm not a cop 🇮🇪)

More: ft.com/content/ba6077...

[Show this thread](#)

People in the UK and US are still going out more than those in locked-down European countries

Google mobility index, change from baseline level



12

141

234

↑



John Burn-Murdoch @jburnmurdoch · Apr 5, 2020

Oh and almost forgot, today's chart-fuel was spaghetti alla ragu. Best ragu we've made yet 🍝



72

58

1.2K

↑

Bad Graphs Collections on the Web

- <https://qz.com/1872980/how-bad-covid-19-data-visualizations-mislead-the-public/>
- <https://venngage.com/blog/misleading-graphs/#Misleading-Coronavirus-graphs>
- <http://www.businessinsider.com/the-27-worst-charts-of-all-time-2013-6>
- https://www.biostat.wisc.edu/~kbroman/topten_worstgraphs/
- <https://www.buzzfeednews.com/article/katienotopoulos/graphs-that-lied-to-us>
- <https://www.statisticshowto.datasciencecentral.com/misleading-graphs/>
- <https://www.reddit.com/r/dataisugly>
- <https://getdolphins.com/blog/the-worst-graphs-of-2017/>
- <https://getdolphins.com/blog/worst-graphs-of-2018/>
- http://bcuchta.com/wall_of_shame/
- ...

Sources for Constructing Better Graphs

Sources for Constructing Better Graphs: Books & Articles

- Robbins, N. B. (2013). Creating More Effective Graphs. Chart House: Ramsey, NJ. (reprinted from Wiley 2005)
- Su, Y-S. (2008). “It’s Easy to Produce Chartjunk using Microsoft Excel 2007, but Hard to Make Good Graphs.” Computational Statistics and Data Analysis 52:4594-4601.
- Tufte, E. R. (2001). The Visual Display of Quantitative Information, 2nd edition. Graphics Press: Cheshire, CT. (first edition 1983)
- Wainer, H. (2000). Visual Revelations: Graphical Tales of Fate and Deception from Napoleon Bonaparte to Ross Perot. Psychology Press: London, UK. (reprinted from Copernicus 1997)

Sources for Constructing Better Graphs:

Blogs

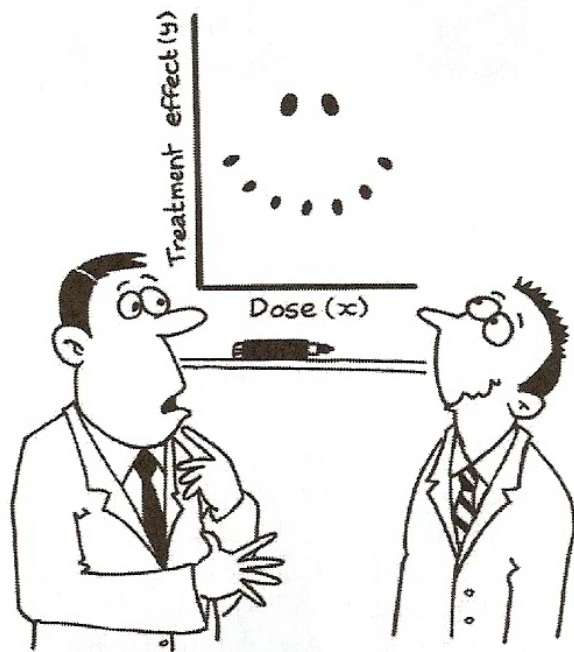
- Camoes, J. “ExcelCharts,” <http://www.excelcharts.com/blog/>
- Peltier, J. “Peltier Tech Blog— Peltier Tech Excel Charts and Programming Blog,” <http://peltiertech.com/WordPress/>
- Robbins, N. B. “Effective Graphs,” <http://www.forbes.com/sites/naomirobbins/>
- Robbins, N. B. (February 28, 2012). “Misleading Graphs: Displaying a Change in One Variable Using Area or Volume,” Forbes.com LLC. <http://www.forbes.com/sites/naomirobbins/2012/02/28/misleading-graphs-displaying-a-change-in-one-variable-using-area-or-volume/>
- Wyatt, S. (March 31, 2014). “Dishonest Fox Charts: Obamacare Enrollment Edition,” Media Matters for America. <http://mediamatters.org/blog/2014/03/31/dishonest-fox-charts-obamacare-enrollment-edition/198679>

Sources for Constructing Better Graphs: For Teachers

- Heiberger, R. M., Robbins, N. B., Symanzik, J. (2014). “Statistical Graphics Recommendations for the ASA/NCTM Annual Poster Competition and Project Competition”, 2014 JSM Proceedings, American Statistical Association, Alexandria, VA.
- Symanzik, J., Robbins, N. B., Heiberger, R. M. (2014). “Observations from the Winners of the 2013 Statistics Poster Competition — Praise and Future Improvements.” The Statistics Teacher Network, 83:2–5.
<http://www.amstat.org/education/stn/pdfs/STN83.pdf>
- Symanzik, J., Robbins, N. B., Heiberger, R. M. (2016): “Observations on the Type and Quality of Graphs Used in the ASA/NCTM Annual Poster Competition during the Years 2013 to 2016”, 2016 JSM Proceedings, American Statistical Association, Alexandria, Virginia.

Acknowledgements

Thanks to my current and former students, my colleagues, and my collaborators who contributed to the collection of bad graphs and bad (and good) graph web pages shown in this presentation. I really appreciate obtaining links to bad graphs and bad graphs web pages even when not discussing bad graphs in the classroom or in a committee or task force. Thank you!



"It's a non-linear pattern with outliers.....but for some reason I'm very happy with the data."

A CAUSE-commissioned cartoon that is part of the CAUSEweb collection and available for free noncommercial use by statistics teachers. Cartoon by John Landers ©. Provided by permission.

Questions ???

— or —

send e-mail to:

symanzik@math.usu.edu

From: Amstat News, January 2009, p. 25